



1 Article

2 An Adaptive Cross-Domain Recommendation Framework for 3 Structurally Disjoint Domains via Semantic Representation 4 and Similar-User Knowledge Transfer

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7 Korea; 20210858@dongduk.ac.kr; 20221829@dongduk.ac.kr; heekim@dongduk.ac.kr8 * Correspondence: heekim@dongduk.ac.kr9 **Abstract**

10 This study addresses cross-domain recommendation when source-domain users and tar-
11 get-domain items are structurally disjoint. We propose an adaptive framework that em-
12 beds source behavior and target-item information in a shared semantic space. Its shared
13 core combines semantic similarity, keyword mapping, preference compatibility, similar-
14 user retrieval, and behavior transfer, while a domain-aware adaptive layer activates sce-
15 nario-specific signals only when the user supplies the corresponding preference. Post-
16 ranking explanations are generated by a separate language model. The framework was
17 evaluated in hobby-to-volunteer and consumption-to-travel scenarios using human and
18 LLM-based relevance judgments and eight baselines, including content-based hybrids
19 with and without transfer and a direct LLM ranker. In Case A, the revised cold-start score
20 outperformed the earlier content hybrid with transfer (Precision@10: 0.420 vs. 0.258;
21 Holm-adjusted $p < 0.001$; Cohen's $d_z = 0.90$) but did not differ significantly from semantic
22 similarity alone. The transfer term did not yield a statistically detectable incremental rank-
23 ing gain in either scenario. In Case B, a two-by-two factorial analysis identified the adap-
24 tive layer as the main source of improvement (+0.363 Precision@10; $p < 0.001$; $d_z = 1.04$),
25 while a control without the adaptive layer showed that part of the association between
26 active signals and performance reflected profile specificity. An exploratory Case A control
27 found no significant explanation-associated change in automated scores. Human–LLM
28 agreement was positive but limited, supporting automated evaluation as a scalable com-
29 plement rather than a replacement for human judgment.

30 **Keywords:** Cross-Domain Recommendation; Structurally Disjoint Domains; Cold-Start
31 Recommendation; Semantic Representation; Similar-User Knowledge Transfer; Adaptive
32 Recommendation Framework; LLM-Based Explanation; LLM-Based Evaluation

33 34 1. Introduction

35 Recommender systems play a central role in helping users discover relevant items
36 from large-scale digital information [1,2]. However, many real-world recommendation
37 scenarios still suffer from data sparsity and cold-start problems, particularly when users
38 have limited interaction histories or target-domain items have not yet been observed [3–
39 5]. Cross-domain recommendation has therefore attracted increasing attention because it

can leverage information from auxiliary domains to improve recommendation quality, alleviate sparsity, and expand the diversity and applicability of recommendations [6,7].

Despite these advantages, most existing cross-domain recommendation methods rely on some form of structural commonality between the source and target domains, such as shared users, shared items, overlapping attributes, or comparable interaction structures [6,7]. User embedding transfer and attribute mapping methods are effective when such cross-domain connections are available and when the two domains exhibit similar data structures [8]. However, these assumptions are often unrealistic in heterogeneous recommendation environments, where the source and target domains may differ fundamentally in purpose, context, and user behavior. For example, hobby activities and volunteer opportunities, or consumption patterns and travel destinations, may be semantically related even though they do not share users, items, or interaction structures. In such settings, direct structural alignment or conventional embedding transfer is insufficient to capture meaningful cross-domain relationships.

The cold-start problem becomes particularly challenging in structurally disjoint environments. New users often lack sufficient interaction histories, preventing conventional collaborative filtering or latent factor-based methods from inferring reliable preferences [3,4]. Recent studies have addressed cross-domain cold-start recommendation through representation learning, domain adaptation, meta-learning-based preference transfer, unsupervised alignment, and similar-user bridging [9–15]. Nevertheless, many of these methods still assume transferable latent structures, bridgeable preference spaces, or sufficient behavioral similarity between domains. When the domains are both semantically heterogeneous and structurally disconnected, recommendation requires a more flexible mechanism that can infer preferences from textual behavior and transfer useful knowledge without explicit cross-domain overlap.

To address these limitations, this study proposes an adaptive cross-domain recommendation framework for structurally disjoint domains. Rather than relying on shared users, shared items, or unified interaction structures, the framework converts source-domain user behavior and target-domain item information as textual representations and projects them into a shared semantic space using sentence-level semantic embeddings [16]. It consists of a shared core recommendation layer and a domain-aware adaptive module. The shared core layer combines semantic similarity, keyword-based mapping, preference compatibility, and similar-user knowledge retrieval. The adaptive module optionally incorporates scenario-specific signals when additional contextual information is available. This design preserves a reusable recommendation architecture while allowing the framework to accommodate the characteristics of different source–target domain pairs [2, 17].

An earlier study by the authors [18] examined hobby-to-volunteer recommendation using textual semantic mapping and similar-user transfer. Table 1 distinguishes the components retained from that study from those introduced or revised here. The present work generalizes the model to a reusable architecture, revises the retrieval and transfer formulation, adds an independent case study and an adaptive layer, and expands the baseline and evaluation protocols.

Table 1. Component-level comparison between the previous study [18] and the present study.

Component	Previous study [18]	Present study
Scenarios	One (hobby to volunteer)	Two (hobby to volunteer; consumption to travel)
Semantic representation	SBERT, Korean only	Language-matched SBERT per case (Korean / English)
Existing-user scoring	$S_t + S_k + S_p$ (0.7 / 0.2 / 0.1)	Inherited unchanged, with per-user percentile normalization
Keyword mapping rules	Hand-written	LLM-generated with author review, in both cases
Similar-user retrieval	TF-IDF cosine similarity	SBERT embedding cosine similarity

Behavior transfer	Unweighted mean of neighbor scores	Similarity-weighted average normalized by the sum of weights
New-user scoring	$0.3S_b + 0.7(S_t + S_k + S_p)$	$w1pct(S_t) + w2pct(S_b)$, weights selected per case
Neighbor-set size K	50	15, selected by a pre-ranking diagnostic scan
Domain-aware adaptive layer	Not present	Per-user activation of lifestyle, attribute and budget signals
Baselines	None	Eight, including random, TF-IDF, SBERT-only and an LLM ranker
Evaluation	User survey ($n = 20$), category match	Human and LLM relevance protocol, five-level rubric
Statistical analysis	None	Bootstrap CIs, paired permutation tests, Holm correction, effect sizes
Explanation-related evaluation bias	Not examined	Exploratory with/without-explanation comparison for the automated evaluator in Case A
Human-LLM agreement	Not examined	Spearman, MAE, quadratic weighted kappa

Large language models (LLMs) have recently been explored in recommender systems because of their semantic understanding, reasoning, explanation generation, and user simulation capabilities [19–24]. Recent studies have applied LLMs to conversational recommendation [25], explainable recommendation [26], and automated recommendation evaluation [27,28]. In the present study, language models are used in five clearly separated roles: synthetic user-profile generation for the evaluation protocol, design-time generation of keyword-mapping rules, post-ranking recommendation explanation, a direct LLM ranking baseline used as an external comparator, and relevance evaluation. None of these language models participates in the run-time score computation of the proposed framework itself; the ranking baseline is evaluated separately as a comparator. Different models are used for generation and evaluation to reduce self-evaluation bias, and agreement between automated and human relevance judgments is assessed explicitly.

The framework is evaluated in two heterogeneous cross-domain recommendation scenarios. Case A uses hobby activity data and volunteer opportunity data to examine whether the shared core layer can generate relevant recommendations without additional domain-specific signals. Case B introduces a new consumption-to-travel scenario to evaluate the generalizability of the shared core layer and the effectiveness of the domain-aware adaptive module. Case A additionally serves as the controlled baseline study: the proposed framework is compared against random ordering, keyword mapping alone, preference compatibility alone, lexical similarity, semantic similarity alone, a content-based hybrid with and without transfer, and a language model used directly as a ranker, all on the same candidate pool, the same profiles, the same list length and the same relevance rubric. In Case B the contribution of the adaptive layer is examined at the level of the individual signals, using the profiles' signal-activation patterns as a factorial design and a model without an adaptive layer as a control.

The main contributions of this study are as follows.

- First, we propose a shared cross-domain recommendation framework that operates across structurally disjoint domains without requiring shared users, shared items, or unified interaction structures. The shared core layer integrates semantic representation, keyword-based mapping, preference compatibility, similar-user retrieval, and behavior transfer.
- Second, we reformulate and empirically evaluate the similar-user transfer mechanism introduced in our previous study. The revised mechanism replaces TF-IDF-based neighbor retrieval with sentence-embedding retrieval and computes the behavior-transfer score as a similarity-weighted average normalized

by the sum of similarity weights. The transfer term introduces information from the existing-user population into the cold-start ranking. However, it did not produce a statistically detectable incremental ranking improvement in either evaluated domain pair, and the conditions under which it improves ranking remain to be established.

- Third, we design and empirically evaluate a domain-aware adaptive module that extends the shared core layer with domain-specific recommendation signals. In the revised formulation this module no longer applies fixed signal weights to every user; each domain-specific signal is activated only for users who supply the corresponding preference, which removes four manually chosen constants from the model and makes the contribution of the individual signals separable. Using a model without an adaptive layer as a control, we show that part of the apparent gain is attributable to profile specificity and quantify the remainder as the contribution of the layer itself.
- Finally, we develop an LLM-supported explanation and evaluation strategy for heterogeneous cross-domain recommendation. The LLM generates evidence-grounded recommendation explanations and supports scalable relevance-based evaluation, while its consistency with human judgment is quantitatively assessed using complementary agreement metrics. In addition, the framework is compared against a panel of competitive baselines under identical experimental conditions, including a large language model used directly as a ranker, and every reported comparison is accompanied by confidence intervals, standardized effect sizes and multiplicity correction.

The remainder of this paper is organized as follows. Section 2 reviews related work on cross-domain recommendation, cold-start recommendation, hybrid recommendation, and LLM-based recommendation systems. Section 3 presents the proposed adaptive cross-domain recommendation framework, including both the shared core recommendation layer and the domain-aware adaptive layer. Section 4 describes the experimental setup and datasets. Section 5 reports experimental results and analysis. Finally, Section 6 concludes the paper and discusses future research directions.

2. Related Work

Cross-domain recommendation aims to improve recommendation performance by transferring user preferences, item knowledge, or behavioral patterns from a source domain to a target domain [5–7, 29]. Existing approaches can be broadly categorized into user-overlap-based, item- or attribute-overlap-based, and representation-transfer-based methods. User-overlap-based methods exploit common or partially shared users to guide preference transfer [6,7]. Item- and attribute-based approaches rely on shared or comparable item characteristics, including categories, popularity, semantic attributes, and textual side information [30,31]. Representation-transfer approaches learn domain-shared user or item embeddings and transfer them across domains [8,11]. These methods have achieved promising results when the source and target domains are structurally connected or partially aligned. However, their applicability is limited when users, items, and interaction structures are entirely disjoint. Because relationships between heterogeneous domains may involve both implicit semantic associations and explicit attribute-level correspondences, integrating multiple recommendation signals can improve robustness when no single signal sufficiently captures cross-domain relevance [2,17].

Cold-start recommendation is closely related to structurally disjoint cross-domain recommendation because new users or target-domain items often lack sufficient interaction histories. Traditional studies have addressed this problem using side information,

latent factor models, and representation learning [3,4]. More recent work has explored meta-learning-based preference transfer, unsupervised alignment for non-overlapping users, heterogeneous domain adaptation, and similar-user bridging [9,10,12–15]. For example, Liu et al. proposed a meta-adversarial framework for transferring source-domain preference representations under cold-start conditions [9], while Wang et al. investigated unsupervised alignment for users observed only in the source domain [10]. Zhou et al. further introduced a similar-user bridging strategy to construct cross-domain connections when overlapping users are limited or unavailable [12]. Although these studies reduce dependence on direct user overlap, many still assume transferable latent structures or bridgeable preference spaces. This assumption may not hold when the source and target domains differ substantially in purpose, semantics, and behavioral context.

Large language models have recently expanded the range of methods available for recommender systems through semantic understanding, natural-language reasoning, explanation generation, and user simulation [19,20,32]. LLMs have been studied as zero-shot conversational recommenders [25], direct cross-domain recommenders [21,33], explanation generators [26], user simulators [22–24], and data augmenters for cold-start items [34,35]. Recent work has also introduced narrative user profiling and dual-channel reasoning for adaptive personalized recommendation [36]. These studies demonstrate that LLMs can support both recommendation generation and auxiliary functions such as profile construction, explanation, and simulation. Recent surveys of generative recommendation models describe how the capabilities listed above are being consolidated into recommendation architectures in their own right rather than used only as auxiliary components [37]. Comparable multimodal models are also being applied to assessment tasks outside recommendation, for example occupational physical exposure assessment from video, which indicates the breadth of settings in which these models are now deployed [38].

LLM-based evaluation has also emerged as a scalable complement to human assessment, particularly in settings where explicit ground-truth labels are unavailable. Zheng et al. examined the agreement between LLM-based judgments and human preferences while identifying potential evaluator biases [27,39]. Liu et al. proposed G-EVAL, which uses structured prompting and probability-based scoring to improve alignment with human judgments [28]. These studies support the use of LLM evaluators for relevance assessment, but they also show that automated evaluation should be interpreted cautiously and validated against human judgment.

Several recent studies are especially relevant to recommendation across weakly connected or disjoint domains. Mayo et al. proposed SPHERE, which constructs semantic personas and retrieves behaviorally similar source-domain communities for recommendation without shared users or items [40]. Adaptive cross-domain methods have also attempted to select source domains or adjust recommendation models according to scenario-specific characteristics [41]. However, these approaches generally focus on semantic persona transfer, source-domain selection, or adaptation of individual models. Comparatively less attention has been paid to reusable architectures that preserve a common recommendation mechanism while selectively incorporating domain-specific signals.

In summary, prior research has advanced cross-domain representation transfer, cold-start mitigation, hybrid recommendation, LLM-supported recommendation, and automated evaluation. Nevertheless, few studies address recommendation between domains that are simultaneously structurally disjoint and semantically heterogeneous while also providing a reusable architecture that supports both general cross-domain transfer and scenario-specific adaptation. The present study addresses this gap through a shared core recommendation layer, a domain-aware adaptive module, and LLM-supported explanation and relevance evaluation.

3. Proposed Adaptive Cross-Domain Recommendation Framework

3.1 Framework Overview

This study proposes an adaptive cross-domain recommendation framework designed for heterogeneous environments in which no structural commonality exists between source-domain users and target-domain items. Unlike conventional cross-domain recommendation approaches that rely on shared users, shared items, or similar data structures, the proposed framework focuses on semantic relationships and transferable behavioral patterns across domains. As shown in Figure 1, the framework consists of two major components: a shared core recommendation layer and a domain-aware adaptive layer.

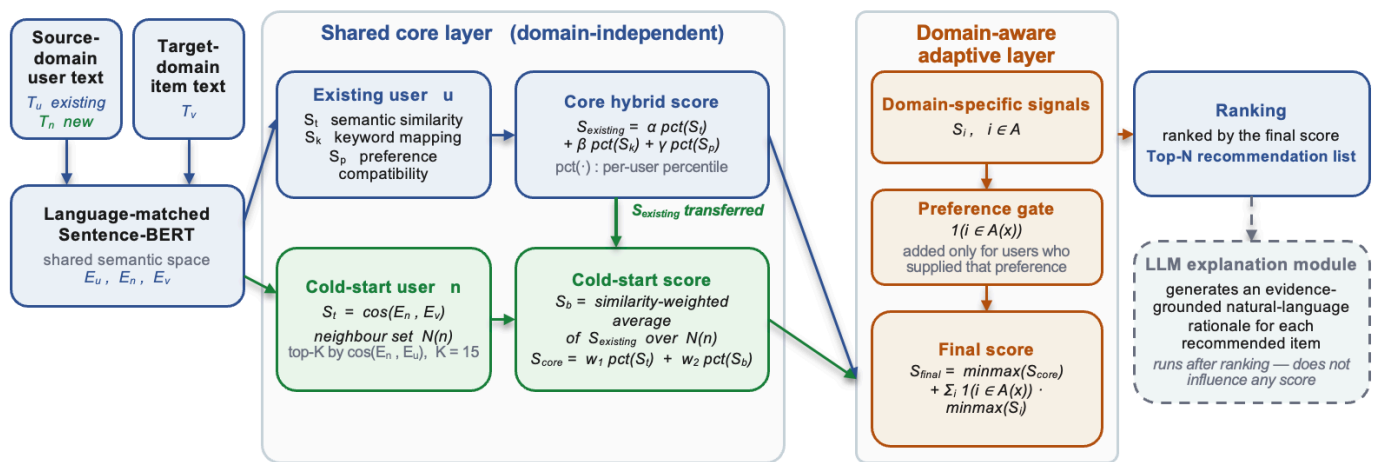


Figure 1. Overall architecture of the proposed adaptive cross-domain recommendation framework. Source-domain user text and target-domain item text are embedded into a shared semantic space using a language-matched Sentence-BERT model. The shared core layer computes a hybrid score for existing users and transfers it to cold-start users through a similarity-weighted aggregation over the $K = 15$ most similar existing users, so that the keyword-mapping and preference-compatibility signals reach cold-start users indirectly rather than being applied to them directly. The domain-aware adaptive layer adds a domain-specific signal only for users who supplied the corresponding preference. The LLM explanation module operates after ranking and does not influence any score.

Language models are used in five clearly separated roles: synthetic user-profile generation, keyword-rule generation, post-ranking explanation generation, direct LLM ranking, and relevance evaluation. None participates in the score computation of the proposed framework; the direct LLM ranker is evaluated separately as a comparator. The five roles are filled by three different models so that the evaluator assesses no text that it generated itself.

The shared core recommendation layer provides a reusable recommendation mechanism that can be applied across different domain pairs. First, source-domain user behavior information and target-domain item information are converted into textual representations and projected into a shared semantic space using SBERT-based semantic embeddings. This process enables the framework to capture latent semantic relationships between heterogeneous domains, even when no explicit structural links exist between users and items. Based on these semantic representations, the core layer computes multiple recommendation signals, including semantic similarity, keyword-based mapping, and preference compatibility. These signals are then integrated through a core hybrid scoring module to generate a basic recommendation score. Before the signals are combined, each

of them is converted to a per-user percentile rank over the candidate set, so that signals with very different raw distributions enter the weighted sum on a common scale; the normalization operators are defined in Equation (4).

For new users, the framework additionally performs similar-user retrieval and behavior transfer. Direct semantic similarity already provides a ranking based on the new-user profile and target-item text, while the transferred signal allows the ranking to incorporate information derived from the existing-user population. In the revised framework, this comparison is performed using sentence embeddings rather than lexical features. The neighbor-set size was fixed at $K = 15$ as a conservative cross-case heuristic based on the Case B coverage diagnostic reported in Supplementary Method S1.4; it was not independently validated in Case A.

Although the shared core layer provides a reusable recommendation mechanism, different recommendation scenarios may require additional domain-specific information that cannot be fully captured by semantic similarity and behavior transfer alone. Therefore, the proposed framework introduces a domain-aware adaptive layer. This layer is designed as an extensible module that incorporates domain-specific features, matching strategies, or scoring components when such information is available. For example, in a consumption-to-travel recommendation scenario, the adaptive layer can use travel-specific signals such as lifestyle traits, destination attributes, and budget compatibility. In this way, the proposed framework maintains a common recommendation architecture while flexibly adapting to the characteristics of individual source and target domains. In the revised formulation this layer is gated per user: a domain-specific signal contributes to the final score only when the corresponding preference has actually been supplied, and a user who supplies none falls back exactly to the core score.

Finally, the ranked recommendation results are passed to an LLM-based explanation module. The LLM does not directly participate in recommendation scoring; instead, it generates natural-language explanations based on the user profile, recommended item information, and supporting recommendation evidence. This module improves the interpretability and transparency of the recommendation results, especially in heterogeneous cross-domain settings where users may not immediately understand why an item from a different domain has been recommended.

3.2 Shared Core Recommendation Layer

The shared core recommendation layer is designed to generate a common set of recommendation signals from heterogeneous source and target domains. It consists of two sequential steps: semantic representation and core hybrid scoring. First, source-domain user information and target-domain item information are converted into unified textual representations and embedded into a shared semantic space using SBERT. Second, multiple recommendation signals, including semantic similarity, keyword-based mapping, and preference compatibility, are integrated to compute the core recommendation score. The sentence-embedding model is selected to match the language of each case study, and no translation step is applied at any point.

Figure 2 shows a two-dimensional UMAP projection of the SBERT-based embeddings used in Case B. The source-domain Amazon user profiles are broadly distributed, whereas the target-domain travel city representations form a more compact region in the projected semantic space. This pattern indicates that the two domains have different semantic distributions, reflecting the difference between consumption-oriented user profiles and travel-oriented destination attributes. Although both types of data are represented using the same embedding model, their semantic characteristics remain domain-specific.

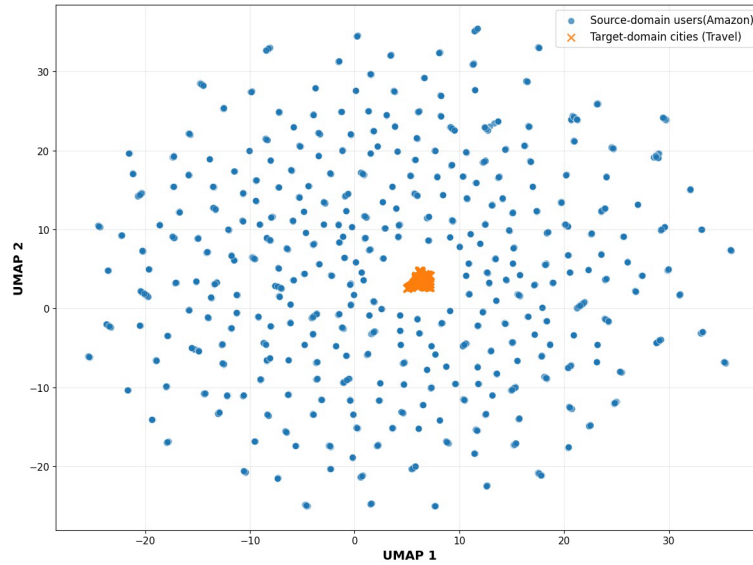


Figure 2. Two-dimensional UMAP projection of SBERT-based semantic embeddings for source-domain Amazon users and target-domain travel cities in Case B. The projection is used for visualization only; the projected distances are not the distances used by the recommendation model, which operates on the full-dimensional embeddings.

The projection shows that the two domains occupy different semantic distributions, which motivates the use of matching and adaptive signals beyond direct lexical overlap. The semantic representation and core-scoring components extend our earlier hobby-to-volunteer model [18] into a shared architecture for heterogeneous domain pairs.

Let T_u , T_n , and T_v denote the textual representations of an existing user, a new user, and a target-domain item, respectively. Before embedding, each input text was processed using a common normalization pipeline consisting of lower-casing, URL removal, removal of non-alphanumeric characters, and whitespace normalization. The normalized texts were then converted into dense sentence-level embeddings using language-matched Sentence-BERT models, enabling semantic relationships to be captured beyond direct lexical overlap. Case A, in which all source- and target-domain texts are Korean, used `jhgan/ko-sroberta-multitask`, whereas Case B, in which the product-review and destination texts are English, used `sentence-transformers/all-mpnet-base-v2`. Within each case, the same embedding model was applied to existing-user representations, new-user profiles, and target-item texts, ensuring that all cosine-similarity computations were performed in a single semantic space. Additional implementation details are provided in Supplementary Method S1. The embeddings are defined as follows:

$$E_u = \text{SBERT}(T_u), E_n = \text{SBERT}(T_n), E_v = \text{SBERT}(T_v) \quad (1)$$

where E_u , E_n , and E_v denote the semantic embedding vectors of existing users, new users, and target-domain items, respectively. Based on these embedding vectors, semantic similarity between a source-domain user and a target-domain item is computed using cosine similarity. For an existing user u and a target-domain item v , the semantic similarity score is defined as follows:

$$S_t(u, v) = \cos(E_u, E_v) \quad (2)$$

For a new user n , the semantic similarity with target-domain item v is computed in the same manner:

$$S_t(n, v) = \cos(E_n, E_v) \quad (3)$$

The semantic similarity score serves as a domain-independent recommendation signal that captures contextual relevance between source-domain user preferences and target-domain item characteristics. This semantic representation forms the foundation of the proposed core recommendation layer and provides the semantic similarity signal used in the hybrid recommendation model. Because the component signals defined below have markedly different raw distributions, each signal is normalized separately for each user over the candidate set V before the signals are combined. Let x denote either an existing user u or a new user n . The percentile-rank and min-max normalization operators are defined as follows:

$$\begin{aligned} [\text{pct}_x(S)]_v &= \begin{cases} \frac{\text{rank}_x(S_{x,v})}{|V| - 1}, & |V| > 1, \\ 0, & |V| = 1, \end{cases} \\ [\text{minmax}_x(S)]_v &= \begin{cases} \frac{S_{x,v} - m_x(S)}{M_x(S) - m_x(S)}, & M_x(S) > m_x(S), \\ 0, & M_x(S) = m_x(S). \end{cases} \end{aligned} \quad (4)$$

Here, $\text{rank}_x(S_{x,v})$ denotes the zero-based ascending rank of $S_{x,v}$ among the candidate items for user x , with tied values assigned the average of their occupied ranks. Further, $m_x(S) = \min_{v' \in V} S_{x,v'}$ and $M_x(S) = \max_{v' \in V} S_{x,v'}$. When the candidate set contains only one item, the percentile-rank value is defined as 0. When all candidate items have the same value for a signal, the min-max normalized value is also defined as 0 because that signal contains no within-user ranking information. Both operators are therefore well defined and bounded in $[0,1]$.

The hybrid recommendation model integrates multiple recommendation signals to compute the core recommendation score for existing users. In heterogeneous cross-domain settings, semantic similarity alone may not fully capture recommendation relevance because user preferences and target-domain item characteristics can be related through both implicit semantic associations and explicit attribute-level matches. Therefore, the proposed model combines three types of signals: semantic similarity, keyword-based mapping, and preference compatibility.

First, the semantic similarity score $S_t(u, v)$ is computed using the SBERT embeddings defined in Section 3.2. This score captures the contextual relevance between the textual representation of source-domain user u and target-domain item v .

Second, the keyword-based mapping score $S_k(u, v)$ is introduced to reflect explicit keyword correspondences between source-domain user interests and target-domain item characteristics. This score is defined as: In the revised framework these mapping rules are no longer written by hand. Candidate keywords are first extracted from the source-domain and target-domain corpora, and a language model operating with deterministic decoding then proposes the semantic correspondences together with a weight for each rule; the generated table is subsequently reviewed by the authors and the reviewed table is the one used. The same procedure was applied to both case studies. This procedure reduces manual rule authoring but does not eliminate human intervention, because the generated rule tables are reviewed before use. In addition, the case-specific preference-compatibility definitions remain manually specified, as documented in the Supplementary Material.

$$S_k(u, v) = \begin{cases} \frac{1}{m} \sum_{i=1}^m w_i I_i, & m > 0 \\ 0, & m = 0 \end{cases} \quad (5)$$

where $w_i \in [0,1]$ is the predefined weight of the i -th mapping rule, I_i equals 1 when the target-domain item matches the keyword pattern associated with that rule and 0 otherwise, and m is the number of mapping rules activated by the user profile. Thus, $S_k(u, v)$ represents the average weighted match score across the activated rules. If no rule is activated, the score is set to 0. The complete case-specific keyword-mapping rules and weights are provided in Supplementary Methods S4.1 and S4.3. The complete generated and reviewed rule tables for both cases, together with the generating model and its decoding settings, are provided in the Supplementary Material.

Third, the preference compatibility score $S_p(u, v)$ is used to capture the alignment between user preferences and target-domain item attributes. The preference-compatibility score is defined in Equation (6) as a bilinear agreement between a behavioral-tendency indicator derived from the user's source-domain behavior and a normalized attribute of the target-domain item.

$$S_p(u, v) = g(u) \cdot a(v) + (1 - g(u)) \cdot (1 - a(v)), \quad g(u), a(v) \in [0,1] \quad (6)$$

where $g(u)$ is a behavioral-tendency indicator derived from the source-domain behavior of user u and $a(v)$ is the corresponding normalized attribute of the target-domain item v , both taking values in $[0,1]$. The score reaches its maximum when the two agree at either extreme and takes the value 0.5 when the user expresses no clear tendency. The interpretation of $g(u)$ and $a(v)$ is case-specific. In Case A, $g(u)$ is the tendency of the user towards popular, large-scale activities and $a(v)$ is the percentile-ranked recruitment capacity of the volunteer activity. In Case B, preference compatibility is instantiated differently: a short textual description of the user's price sensitivity, derived from the average discount rate of the products the user purchased, is compared with a textual description of the price positioning of each destination using the same sentence-embedding model, and the resulting cosine similarity is rescaled to $[0,1]$. Both instantiations are documented in the Supplementary Material.

Finally, the core recommendation score is computed as a weighted combination of the three components:

$$S_{\text{existing}}(u, v) = \alpha \cdot \text{pct}(S_t(u, v)) + \beta \cdot \text{pct}(S_k(u, v)) + \gamma \cdot \text{pct}(S_p(u, v)) \quad (7)$$

where α , β , and γ are weighting parameters that control the relative contribution of semantic similarity, keyword-based mapping, and preference compatibility, respectively. In both cases, the weights were set to $\alpha = 0.7$, $\beta = 0.2$, and $\gamma = 0.1$. The complete scoring and hyperparameter settings are summarized in Supplementary Table S4. Based on the computed scores, candidate target-domain items are ranked, and the top- N items are selected as the final recommendations for the user. Each component score entering this combination is first converted to a per-user percentile rank as defined in Equation (4), so that the weights can be interpreted as relative contributions rather than being confounded with differences in the scale of the individual signals.

Algorithm 1 summarizes the hybrid recommendation procedure for existing users.

Algorithm 1. Hybrid Cross-Domain Recommendation for Existing Users

Input: existing user u , target-domain item set V , item texts $\{T_v\}$

Output: Top- N recommended items R

-
1. Construct the source-domain user text T_u by aggregating available behavioral information.
 2. Compute the user embedding $E_u = SBERT_c(T_u)$, where c is the language-matched model of the case study.
 3. For each target-domain item $v \in V$:
 - a. Compute the item embedding $E_v = SBERT_c(T_v)$.
 - b. Compute the semantic similarity $S_t(u, v) = \cos(E_u, E_v)$.
 - c. Compute the keyword-based mapping score $S_k(u, v)$ using the generated and reviewed rule table
 - d. Compute the preference compatibility score $S_p(u, v)$.
 4. Convert S_t , S_k and S_p to per-user percentile ranks $pct(\cdot)$ over the candidate set V .
 5. Compute the core score $S_{existing}(u, v) = \alpha \cdot pct(S_t) + \beta \cdot pct(S_k) + \gamma \cdot pct(S_p)$, with $(\alpha, \beta, \gamma) = (0.7, 0.2, 0.1)$.
 6. Rank all items in V according to $S_{existing}(u, v)$.
 7. Return the Top-N items as R .
-

3.3 User Behavior Transfer for Cold-Start Recommendation

To alleviate the cold-start problem for new users, the proposed framework incorporates a similar-user knowledge transfer mechanism. In structurally disjoint cross-domain recommendation, new users often have limited or no interaction histories in the target domain. Therefore, relying only on direct user-item semantic similarity may be insufficient. The framework addresses this limitation by identifying existing users with similar behavioral characteristics and transferring their recommendation signals to the new user. Two aspects of this mechanism were redesigned in the present study on the basis of the additional experiments reported in Section 5. Neighbor retrieval now uses SBERT sentence embeddings instead of TF-IDF features, because lexical overlap between a short cold-start profile and an aggregated existing-user document is sparse and brittle, whereas sentence-level embeddings retrieve behaviorally similar users even when the two texts share few surface terms; this also places neighbor retrieval and user-item matching in the same representation space. The neighbor-set size was fixed at $K = 15$ as a conservative heuristic after complete signal coverage was confirmed at $K = 10$ in Case B. The diagnostic used only neighbor-similarity and signal-coverage statistics, not relevance labels or recommendation-quality metrics, and was not independently repeated in Case A.

Similar-user retrieval is supported by hierarchical user clustering. In the first-level clustering stage, users are grouped according to broad behavioral characteristics derived from domain-specific activity or consumption patterns. This stage captures high-level preference tendencies and narrows the candidate search space. In the second-level clustering stage, user-related textual behavioral information is transformed into TF-IDF representations [42] and further clustered to identify fine-grained preference patterns within each high-level group. UMAP is used for nonlinear dimensionality reduction [43], and HDBSCAN is employed to identify density-based clusters without requiring a predefined number of clusters [44]. The UMAP configuration was kept consistent across both case studies and clustering stages, whereas the HDBSCAN parameters were adjusted accord-

ing to the source-domain population size and clustering stage. Detailed clustering procedures and parameter settings are provided in Supplementary Method S2 and Tables S2 and S3.

Figure 3 illustrates the similar-user knowledge transfer process in Case B. A new source-domain user is compared with existing users on the basis of SBERT embedding similarity, and the 15 most similar existing users ($K = 15$) were selected as the neighbor set for behavior transfer. For visual clarity the figure marks only the most similar neighbors rather than the full neighbor set; the transfer computation in Equation (9) uses all fifteen. The recommendation signals associated with these similar users are then transferred to the new user and combined with the semantic similarity score to generate the final recommendation.

In the figure, one new Amazon user, retrieved similar users, and a recommended travel city are highlighted in the shared semantic space. Although some users may appear visually closer to the new user in the two-dimensional UMAP projection, the actual neighborhood is determined by the similarity computation in the recommendation model rather than by visual proximity alone. This mechanism enables the framework to reuse behavioral knowledge from existing users and improve recommendation robustness under cold-start conditions.

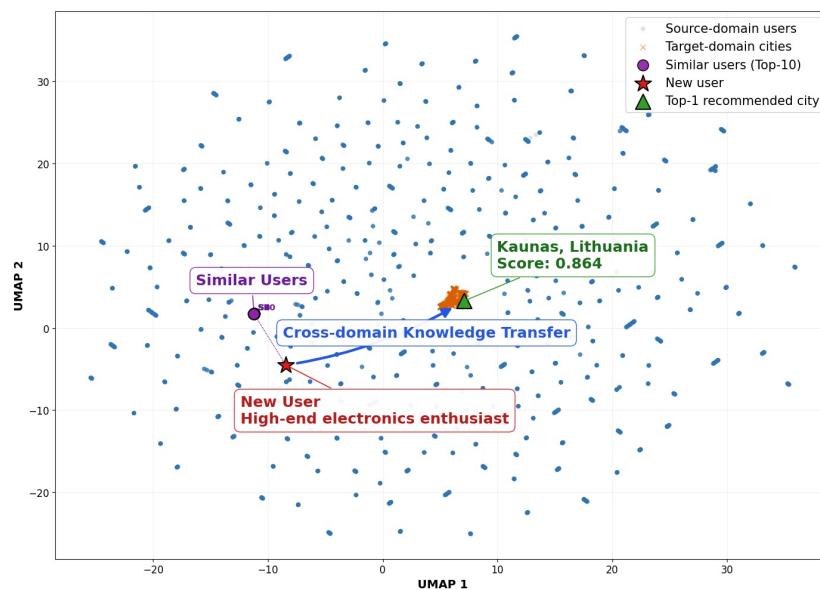


Figure 3. Illustrative similar-user knowledge transfer process for Case B. A new source-domain user is matched with behaviorally similar existing users retrieved by SBERT embedding similarity, and their recommendation signals are transferred to support travel destination recommendation. The figure marks the most similar neighbors only; the behavior-transfer score of Equation (9) is computed over the complete neighbor set of $K = 15$ existing users.

For a new user n , the profile text T_n is embedded with the same sentence-embedding model used for the case study. Cosine similarity is then computed between the new user and the existing users, and the 15 most similar existing users are selected as the neighbor set $N(n)$, as defined in Equation (8). For each target-domain item v , the behavior transfer score is computed as a similarity-weighted average of the core scores of those neighbors that actually hold a score for that item, normalized by the sum of their similarity weights. Negative similarities are clipped to zero before weighting, so that no neighbor can contribute with a negative weight.

$$N(n) = \arg \max_{u \in U} -K \cos(E_n, E_u), \quad N_v(n) = \{u \in N(n) : S_{\text{existing}}(u, v) \text{ is defined}\} \quad (8)$$

$$S_b(n, v) = \begin{cases} \frac{\sum_{u \in N_v(n)} \text{sim}(n, u) \cdot S_{\text{existing}}(u, v)}{\sum_{u \in N_v(n)} \text{sim}(n, u)}, & \text{if } N_v(n) \neq \emptyset \\ 0, & \text{if } N_v(n) = \emptyset \end{cases} \quad (9)$$

where $S_b(n, v)$ denotes the behavior transfer score for new user n and target-domain item v , $\text{sim}(n, u)$ denotes the SBERT cosine similarity between new user n and existing user u , and $S_{\text{existing}}(u, v)$ is the core recommendation score of existing user u for item v . Because every component of S_{existing} is converted to a percentile rank in $[0, 1]$ before combination, and S_b is a convex combination of such values with non-negative weights, S_b is itself bounded in $[0, 1]$ by construction; the transferred signal and the semantic similarity signal therefore enter Equation (10) on an identical scale. In all experiments K was fixed at 15. Additional implementation details are provided in the Supplementary Material.

The submitted cold-start formulation also applied keyword-mapping and preference-compatibility scores directly to the new user. These terms were removed because, without interaction history, they performed at or below the random baseline (Table 5). They remain in the existing-user score and therefore reach the new user indirectly through the behavior-transfer term S_b .

$$S_{\text{core}}(n, v) = w_1 \cdot \text{pct}[S_t(n, v)] + w_2 \cdot \text{pct}[S_b(n, v)], \quad w_1 + w_2 = 1 \quad (10)$$

where w_1 and w_2 are weighting parameters with $w_1 + w_2 = 1$ that control the relative contribution of direct semantic similarity and transferred behavior. The selected values differ between the two case studies, $(w_1, w_2) = (0.9, 0.1)$ in Case A and $(0.6, 0.4)$ in Case B. Both settings were obtained from the weight-sensitivity scan reported in the Supplementary Material, which also includes the setting $(1.0, 0.0)$ in which the transferred term is switched off entirely. The transferred term introduces information from the existing-user population into the cold-start ranking rather than making ranking possible by itself, because direct semantic similarity already provides a defined score. Its incremental effect on the ranking is evaluated in Section 5.2.5. Based on the resulting scores, all candidate target-domain items are ranked and the Top- N items are selected as the final recommendations for the new user.

Algorithm 2. Cold-Start Recommendation via Similar-User Knowledge Transfer

Input: new user profile text T_n , existing user set U , target-domain item set V , item texts $T_{v \in V}$

Output: Top- N recommended items R

1. Embed the new user profile text T_n with the language-matched sentence-embedding model of the case study, giving E_n .
 2. For each existing user $u \in U$, compute the cosine similarity $\text{sim}(n, u)$ between E_n and the embedding of the source-domain text of u , clipping negative values to zero.
 3. Select the K existing users with the highest similarity as the neighbor set $N(n)$; $K = 15$ in the reported configuration.
 4. For each target-domain item $v \in V$:
 - a. Compute the item embedding E_v and the semantic similarity $S_t(n, v) = \cos(E_n, E_v)$.
-

-
- b. Compute $S_b(n, v)$ as the similarity-weighted average of the existing-user scores $S_{existing}(u, v)$ of Equation (7) over the neighbors in $N(n)$ that hold a score for v , normalized by the sum of their similarity weights; $S_b(n, v) = 0$ if no neighbor holds a score.
 5. Convert S_t and S_b to per-user percentile ranks $pct(\cdot)$ over the candidate set V .
 6. Compute the core score $S_{core}(n, v) = w_1 \cdot pct(S_t) + w_2 \cdot pct(S_b)$, with $w_1 + w_2 = 1$; $(w_1, w_2) = (0.9, 0.1)$ in Case A and $(0.6, 0.4)$ in Case B.
 7. Apply min-max normalization to S_{core} over V and add the gated adaptive signals, giving $S_{final}(n, v)$ as defined in Equation (11). If the user supplies no domain-specific preference, S_{final} reduces to the normalized core score.
 8. Rank all items in V by $S_{final}(n, v)$.
 9. Return the Top-N items as R .
-

3.4 Domain-Aware Adaptive Recommendation

Although the shared core recommendation layer provides a common set of recommendation signals, certain recommendation scenarios require additional contextual information that cannot be fully captured by semantic similarity, keyword-based mapping, preference compatibility, or behavior transfer alone. In heterogeneous recommendation environments, domain-specific characteristics often play an important role in determining recommendation relevance. Therefore, the proposed framework introduces a domain-aware adaptive recommendation component that incorporates domain-specific signals while preserving the shared core recommendation architecture. In the submitted version of this manuscript the domain-specific signals were applied to every user with fixed weights. That formulation was revised for two reasons. First, it is conceptually inappropriate: a user who expresses no budget preference should not receive a budget-compatibility score at all. Second, it makes the contribution of the individual signals impossible to separate, because every user receives every signal.

The adaptive recommendation component is designed as an extensible module that can integrate multiple domain-specific recommendation signals. Let A denote the set of available domain-specific signals and $A(x)$ the subset for which user representation x has actually supplied the corresponding preference. Each signal S_i is min-max normalized over the candidate set of the user and is added to the normalized core score only when it is active, as expressed by the indicator $1(i \in A(x))$. This formulation removes the fixed signal weights and the balancing parameter used previously, so that the adaptive layer introduces no additional tuned constants. A user for whom $A(x)$ is empty receives exactly the core score, which we verify programmatically in the implementation.

$$S_{final}(x, v) = \text{minmax}[S_{core}(x, v)] + \sum_{i \in A} \mathbf{1}(i \in A(x)) \cdot \text{minmax}[S_i(x, v)] \quad (11)$$

where $\text{minmax}(\cdot)$ is the per-user min-max normalization operator defined in Equation (4). The specific definition of each signal S_i depends on the target domain and application scenario; the three signals instantiated for the consumption-to-travel case are defined in Equations (12) to (14). Rather than replacing the shared core recommendation layer, the adaptive component extends the ranking score with additional domain-specific information for those users who provide it.

The fixed-weight formulation used in the submitted version, which applied all three adaptive signals to every user through three constants and a balancing parameter, has been replaced by the indicator-gated score of Equation (11).

Here $S_{core}(x, v)$ denotes the score obtained from the shared core layer: for an existing user it is $S_{existing}(u, v)$ of Equation (7), and for a new user it is $S_{core}(n, v)$ of Equation (10). The case-specific definitions of the adaptive signals instantiated in Case B are given in Section 4.2.2 and in the Supplementary Material.

3.5 LLM-based Explanation

To enhance the interpretability of recommendation results, the proposed framework incorporates an LLM-based explanation generation module. The LLM is not used to compute recommendation scores; instead, it serves as an auxiliary component that generates natural-language explanations after the recommendation ranking has been determined. By explaining the relationship between source-domain user preferences and target-domain item characteristics, the module helps users understand why an item from a different domain has been recommended. The model used for explanation generation is different from the model used for relevance evaluation in Section 5.1 and from the model used to generate the synthetic evaluation profiles, so that no model assesses text that it produced itself. The exact models and decoding settings are reported in Section 5.1 and in the Supplementary Material.

The explanation module follows a data-grounded prompting strategy. Specifically, the LLM receives structured inputs consisting of the source-domain user profile, target-domain item information, and supporting recommendation evidence such as semantic similarity, shared keywords, preference compatibility, or domain-specific matching signals. The prompt is designed to guide the LLM to identify meaningful connections between the user and the recommended item while preventing unsupported assumptions. Table 2 presents the structured prompt template used to organize the explanation-generation process. The complete case-specific prompts are provided in Supplementary Method S5.2. The layered structure of this prompt follows evidence from systematic evaluations of prompt design for language-model-based personalized recommendation, which report that the organization of the prompt, in particular the explicit separation of context from reasoning instructions, materially affects output quality [45].

Table 2. Prompt template for LLM-based explanation generation.

Prompt Template
System instruction: You are an assistant that provides clear, concise, and evidence-grounded explanations for recommendation results. The explanation should be based only on the provided information and should not include unsupported assumptions or external knowledge.
[Layer 1: Context] You are given the following information: • User profile (source-domain behavioral information) • Target-domain item information • Supporting recommendation evidence, such as shared keywords, semantic concepts, preference compatibility, domain-specific matching signals, available metadata, or user feedback * Use only the information as the basis for the explanation.
[Layer 2: Reasoning]

Based on the provided context, identify meaningful connections between the user and the target-domain item.

Consider the following evidence:

- semantic relevance between user preferences and item characteristics
- shared keywords or concepts
- compatibility between user preferences and item attributes
- domain-specific matching signals
- available metadata or user feedback, if provided

*Do not introduce any unsupported assumptions.

[Layer 3: Response]

Generate exactly three recommendation points explaining why the target-domain item is suitable for the user.

Output requirements:

- Exactly three sentences
 - Each sentence should be concise
 - At least one sentence must reflect information derived from available feedback or supporting evidence
 - Do not include technical terms such as model, score, algorithm, embedding or semantic similarity
 - Do not include information that is not provided in the input
-

- Note: The prompt was originally executed in Korean because the user interface and human evaluation involved Korean-speaking participants.

The model was instructed to use only the supplied profile, item information, metadata, feedback, and recommendation evidence, reducing reliance on external knowledge and unsupported content [46,47]. In addition, a rule-based fallback mechanism was used to maintain explanation availability when the LLM API call failed or returned an empty response. In such cases, the system generated a template-based explanation derived from the available user profile, target-domain item metadata, and recommendation evidence. The fallback mechanism was intended to ensure stable explanation delivery, while unsupported or irrelevant content was primarily mitigated through data-grounded prompting rather than automatic post-generation verification. Through this combination of structured prompting, data-grounded generation, and error-triggered fallback, the explanation module improves the transparency and interpretability of recommendation results in heterogeneous cross-domain recommendation environments. Detailed model configurations and the case-specific prompts used for recommendation explanation are provided in Supplementary Method S5.2 and Table S10.

4. Experimental Design and Case Study Scenarios

4.1. Case A: Hobby-to-Volunteer Recommendation

4.1.1. Dataset and Text Representation

Case A was designed to examine whether the shared core recommendation framework can generate meaningful recommendations across structurally disjoint domains without using domain-specific adaptive components. In this scenario, hobby activity data

were used as the source domain, and volunteer opportunity data were used as the target domain. The source-domain data were compiled by crawling publicly accessible web pages from FRIP [48], a hobby activity platform, without user authentication. Similarly, the target-domain data were gathered from the Volunteer Management System (VMS) [49] via the same public crawling method. The resulting dataset comprises 1,612 users with 190 hobby items for the source domain, and 1,500 volunteer opportunity listings for the target domain. For the revised experiments, the target-domain pool was reduced from 1,500 to 500 listings by stratified sampling so that all compared methods operated on an identical candidate set within a single context window.

The two domains are structurally disjoint because there are no shared users, shared items, or direct interaction records between hobby activities and volunteer opportunities. Therefore, recommendation cannot rely on conventional user–item interactions or cross-domain overlap. Instead, the proposed framework constructs textual representations from the available information in each domain and uses them as the basis for semantic matching.

For the hobby domain, activity descriptions, user reviews, and activity categories were aggregated to construct source-domain user representations. These elements reflect users' interests, preferred activity types, and behavioral tendencies. For the volunteer domain, activity titles, descriptions, target groups, and location information were used to construct target-domain item representations. These elements describe the characteristics and participation context of each volunteer opportunity.

All text was normalized by removing special characters and unnecessary tokens before input to the shared core framework. The resulting representations placed hobby preferences and volunteer-opportunity characteristics in a common textual form. Case A contained 1,612 existing users, compared with 9,050 in Case B; the clustering settings are reported in Supplementary Tables S2 and S3.

4.1.2 Application of the Shared Core Framework

In Case A, the shared core recommendation framework was directly applied to the hobby-to-volunteer recommendation task. No additional domain-aware adaptive module was introduced in this case, because the purpose of Case A was to evaluate the basic applicability of the proposed core framework in a structurally disjoint cross-domain setting.

For each existing user, hobby-related behavioral information was aggregated into a unified textual representation. For new users, profile information collected through the input form was used to construct a new user representation. Volunteer activities were also converted into textual item representations using their descriptions and metadata. These user and item representations were then processed using the shared core framework described in Section 3.

Specifically, the 15 most similar existing users were retrieved for each new user with SBERT embedding similarity, and their recommendation scores were aggregated using similarity-based weights, normalized by the sum of those weights, to construct the behavior transfer signal.

Hierarchical user clustering was used as an auxiliary step to support similar-user retrieval. Users were first grouped according to broad hobby-category patterns, and more fine-grained preference patterns were then identified within each group using textual behavioral features. Detailed clustering procedures, parameter settings, and visualization results for Case A are provided in Supplementary Method S2.1 and Figures S2 and S3.

Based on the final recommendation scores, candidate volunteer activities were ranked, and the Top-N activities were selected as the final recommendations. After ranking, the LLM-based explanation module generated natural-language explanations using

the user profile, recommended volunteer activity information, and supporting recommendation evidence. Through this process, Case A demonstrates that the shared core framework can support cross-domain recommendation between hobby activities and volunteer opportunities without relying on shared users, shared items, or domain-specific adaptive signals.

4.2. Case B: Consumption-to-Travel Recommendation

Case B evaluates the shared core and domain-aware adaptive layers in a consumption-to-travel setting distinct from the hobby-to-volunteer scenario of our earlier study [18].

4.2.1. Dataset and Text Representation

Case B was designed to evaluate the adaptability of the proposed framework in a heterogeneous recommendation scenario that requires domain-specific recommendation signals. In this case study, product consumption data were used as the source domain, and travel destination data were used as the target domain.

The source-domain data were obtained from the Amazon Sales Dataset [50] available on Kaggle, while the target-domain data were retrieved from a Worldwide Travel cities dataset [51], also sourced from Kaggle. The Amazon dataset encompasses 9,050 users and 1,351 product records across multiple categories, including product descriptions, user reviews, ratings, and category information. These records were utilized to construct source-domain user representations that reflect users' consumption preferences and purchasing behaviors. Meanwhile, the travel dataset for the target domain contains 560 city destination records, including textual descriptions and structured attributes such as budget level, culture, adventure, nature, beaches, nightlife, cuisine, wellness, urban characteristics, and seclusion scores.

The two domains are structurally disjoint because consumption records and travel destinations do not share users, items, or direct interaction histories. Therefore, the recommendation task requires the system to infer latent relationships between consumption behavior and travel preferences. To support this process, Amazon reviews and product information were transformed into source-domain user representations, while city descriptions and travel-related attributes were used to construct target-domain item representations. These representations were then processed through the shared core framework and further refined using the domain-aware adaptive module.

4.2.2 Application of the Adaptive Recommendation Module

Unlike Case A, the consumption-to-travel recommendation scenario requires additional domain-specific information beyond the shared core recommendation signals. Although the shared core framework captures semantic relevance, keyword-level associations, preference compatibility, and similar-user-based behavior transfer, travel recommendation also depends on contextual factors such as lifestyle orientation, destination characteristics, and budget compatibility. Therefore, Case B applies the domain-aware adaptive module described in Section 3.4 to incorporate travel-specific recommendation signals.

In this case study, the adaptive module translates consumption-domain signals into travel-domain preference factors. Consumption patterns were used to infer lifestyle tendencies that may be relevant to travel preferences. For example, users who frequently purchase or review products related to outdoor activities, wellness, digital devices, luxury goods, or lifestyle products may exhibit different travel tendencies. These consumption-derived tendencies were then matched with structured destination attributes in the travel dataset.

The adaptive layer uses three travel-specific signals: lifestyle-trait matching, destination-attribute matching, and budget compatibility. Each signal is activated only when the user supplies the corresponding preference. Their definitions are given in Equations (12)–(14), and the complete mappings and scoring rules are reported in Supplementary Method S4.

For a new user n and a target travel destination v , the three domain-specific signals are defined as follows. Equation (12) defines the lifestyle-trait score, where $L(n)$ is the set of lifestyle concepts supplied by the user, D is the set of nine measured destination attributes, $d(c)$ is the attribute most similar to concept c in the embedding space, and $a_d(v)$ is the value of attribute d for destination v .

$$S_L(n, v) = \frac{1}{|L(n)|} \sum_{c \in L(n)} a_{d(c)}(v), \quad d(c) = \arg \max_{d \in D} \cos(E_c, E_d) \quad (12)$$

Equation (13) defines the destination-attribute score, where $F(n)$ is the set of contextual fields the user actually specified among region, trip duration and temperature band, and $match_f(n, v)$ is 1 for an exact match, 0.5 for an adjacent temperature band, and 0 otherwise. Fields that the user did not specify are excluded from the average rather than contributing a default value.

$$S_C(n, v) = \frac{1}{|F(n)|} \sum_{f \in F(n)} match_f(n, v), \quad F(n) \subseteq \{\text{region, duration, temperature}\} \text{ (fields specified by user } n) \quad (13)$$

Equation (14) defines the budget-compatibility score, where $tier(\cdot)$ maps the three-level price vocabulary of each domain onto a common ordinal scale, so that an exact match scores 1, an adjacent tier scores 0.5, and a two-step difference scores 0. The three normalized signals are then added to the normalized core score exactly as specified by Equation (11), each only when the corresponding preference is present. No balancing parameter and no fixed signal weights are used. The complete parameter settings are summarized in the Supplementary Material.

$$S_B(n, v) = 1 - 0.5 \cdot |tier(n) - tier(v)|, \quad tier(\cdot) \in \{0, 1, 2\} \quad (14)$$

Hierarchical consumer clustering was also used as an auxiliary step to support similar-user retrieval in the consumption domain. Consumers were first grouped according to broad product-category preferences, and fine-grained consumption patterns were then identified within each group using textual behavioral features. Detailed clustering procedures, parameter settings, and visualization results for Case B are provided in Supplementary Method S2.2, Table S3, and Figures S4 and S5.

Based on the final adaptive recommendation scores, candidate travel destinations were ranked, and the Top-N destinations were selected as the final recommendations. After ranking, the LLM-based explanation module generated natural-language explanations describing the relationship between the user's consumption characteristics and the recommended travel destinations. Through this process, Case B demonstrates how the proposed framework can extend the shared core recommendation layer by incorporating domain-specific adaptive signals in a structurally disjoint consumption-to-travel recommendation scenario.

5. Experimental Evaluation

5.1. Evaluation Setup

The proposed framework was evaluated from two perspectives: recommendation quality and the agreement between LLM-based and human evaluation. The evaluation

was conducted using the two heterogeneous cross-domain scenarios described in Section 4. Case A evaluated hobby-to-volunteer recommendation using the shared core layer, whereas Case B evaluated consumption-to-travel recommendation using both the shared core layer and the domain-aware adaptive module.

Unlike conventional recommendation benchmarks, structurally disjoint cross-domain recommendation does not provide explicit ground-truth interactions between source-domain users and target-domain items. In the proposed scenarios, target-domain items were generally unseen by users, and multiple items could reasonably match the same user profile. Therefore, this study adopted a relevance-based evaluation framework that assessed how well each recommended item matched the user's inferred interests and characteristics. Accordingly, recommendation quality was assessed using evaluator-derived relevance labels rather than observed target-domain interactions.

Both LLM-based and human evaluation protocols were employed. For the LLM-based evaluation, 50 synthetic user profiles were generated for each recommendation scenario using claude-sonnet-5. These profiles were used as inputs to the proposed recommendation system to generate Top-10 recommendation lists. To reduce self-evaluation bias, the resulting recommendations were evaluated by a different LLM, gemini-3.8-flash, which was used solely as the evaluator. The profile-generation and evaluation stages were conducted using separate prompts and independent API calls. A third model, gpt-5.6-terra, was used to generate the recommendation explanations and to implement the language model ranking baseline, so that the model that produced the text under assessment was different from the model that assessed it and from the model that produced the evaluation profiles. This model separation was adopted to reduce evaluator self-preference and related judge biases documented in prior LLM-as-a-judge studies, including sensitivity to generated style and response length [27,37].

For the human evaluation, 30 participants provided their own source-domain preference information through the recommendation interface. The group comprised 12 men and 18 women; 20 participants were in their twenties, four in their thirties, four in their forties, and two in their fifties. Participants were recruited from the university student body and through an online survey form, and participation was voluntary and unpaid. Each participant evaluated one Top-10 recommendation list in each case study using the same relevance question and five-point scale applied in the LLM-based evaluation, yielding 300 human relevance judgments per case. Each participant evaluated only the final model configuration for each case study. No participant evaluated both the Core Only and Core + Adaptive configurations for Case B. Items were presented in descending order of the model score rather than in randomized order. Thus, the LLM-based protocol evaluated recommendations generated from synthetic profiles, whereas the human protocol evaluated recommendations generated from real participant profiles.

In the primary automated and human evaluation protocols, each recommended item was presented together with its LLM-generated explanation. To examine whether the explanation influenced automated relevance scoring, an additional exploratory control analysis was conducted for Case A. The same recommendation instances were re-evaluated under the same automated-evaluator settings with the explanation omitted. This control analysis was not conducted for Case B or for the human evaluation, and its results are reported in Section 5.5.

For the human-LLM agreement analysis, automated ratings were additionally obtained for the same recommendation instances evaluated by the participants. Complete paired human-automated ratings were available for 20 of the 30 participants, yielding 200 item-level pairs per case. This subset was used exclusively for the agreement analysis reported in Section 5.4. These paired records represent human and automated ratings of the

same recommendation instances and do not constitute a within-participant comparison between the Core Only and Core + Adaptive configurations.

5.1.1. Synthetic User Profiles

Because the two evaluation protocols differ in the kind of user profile they use, we describe the construction of the synthetic profiles in full and state explicitly how they differ from the input of a real cold-start user. The synthetic profiles make it possible to compare many methods under identical conditions, but they are likely to be more explicit and internally more consistent than free-text input from genuine users; the synthetic-profile results should therefore be read as an upper reference rather than as an estimate of deployed performance.

In Case A, 50 hobby-domain profiles were generated. The profile schema is grounded in a published hobby-topic taxonomy[52] rather than defined ad hoc, and every categorical field is drawn from the option set that actually occurs in the crawled platform data: hobby categories, participation style, indoor or outdoor orientation, preferred volunteer activity types, preferred target groups and preferred regions. Each profile also contains a short free-text description of the user's hobbies and a statement of activities the user avoids.

In Case B, 50 consumption-domain profiles were generated. The generation procedure follows recent work that applies large language models to consumer segmentation and to the construction of synthetic user profiles and queries for tourism recommendation [53-55]. Purchase and interest categories were drawn from the categories present in the source dataset, and the contextual options—region, trip duration, travel month, temperature band, and budget level—were drawn from the values present in the target dataset. As a post-generation diversity diagnostic, the maximum pairwise SBERT similarity of each profile to any other profile was recorded. A threshold of 0.92 was used as a screening flag rather than as an exclusion criterion. Thirty-one profiles exceeded the threshold; all were manually reviewed and retained. The adaptive preference fields were deliberately distributed across the profile set so that all eight activation patterns of the three domain-specific signals were represented.

The released profile files record the generating model, random seed, prompt version, validation status, and maximum pairwise similarity. Table 3 summarizes both profile sets; complete prompts and option lists are provided in the Supplementary Material. The near-duplicate flags were distributed across the zero-, one-, two-, and three-signal groups (7, 7, 9, and 8 profiles), reducing concern that the screening flag confounded the component analysis in Section 5.2.3.

Table 3. Construction of the synthetic user profiles used in the LLM-based evaluation protocol.

Item	Case A (hobby to volunteer)	Case B (consumption to travel)
Number of profiles	50	50
Generating model	claude-sonnet-5	claude-sonnet-5
Random seed	42	42
Free-text fields	hobby description, disliked activities	consumption description
Categorical fields	hobby category, participation style, location style, activity type, target group, region	purchase items, interest items, purchasing criteria, lifestyle, contextual fields, budget level
Source of the option sets	observed distribution of the crawled platform data	observed values of the source and target datasets
Schema grounding	published hobby-topic taxonomy	nine measured attributes of the target dataset
Similarity screening	not applied	Maximum pairwise similarity recorded for each profile; threshold 0.92 used as a post-generation

Adaptive signal allocation	not applicable	screening flag (31 of 50 flagged; all manually reviewed and retained).
Audit log	model, seed, prompt version	13 profiles with none, 13 with one, 12 with two, 12 with all three model, seed, prompt version, validation status, maximum similarity

For each recommendation instance, the evaluation materials comprised the corresponding source-domain user profile, the recommended target-domain item, and the LLM-generated explanation. Recommendation relevance was assessed with the following question:

"How well does this recommendation match the interests and behavioral tendencies that are stated in, or can reasonably be inferred from, the provided user profile?"

To ensure consistency across evaluators, the five-point relevance scale shown in Table 4 was incorporated into the LLM evaluation prompt and provided to the human participants as evaluation guidelines.

Table 4. Relevance scale incorporated into the LLM prompt and provided to human evaluators.

Relevance Level	Meaning
1	Completely irrelevant
2	Weakly relevant
3	Neutral / Partially relevant
4	Relevant
5	Highly relevant

For each recommendation instance the evaluator returns a single integer relevance level on this scale, together with a one-sentence justification. The relevance score is therefore

$$Score_{LLM}(u, v) = r(u, v), r(u, v) \in 1, 2, 3, 4, 5 \quad (15)$$

where $r(u, v)$ denotes the integer level assigned by the evaluator to the recommendation of item v to user u under the rubric of Table 4. The submitted version of this manuscript instead adopted the probability-weighted formulation of G-EVAL [28], in which the evaluator estimates a distribution over the five levels and the score is taken as its expectation. That formulation has been replaced because the human protocol elicits a single categorical rating on the same five-point scale, and a continuous expectation is not commensurable with it: the two score distributions differ in support and in granularity, which affects the agreement statistics reported in Section 5.4 and in particular the quadratic weighted kappa, which is defined over ordinal categories. Under the present formulation $Score_{LLM}$ and $Score_{Human}$ are defined on an identical five-level ordinal scale and are directly comparable without any post-hoc transformation.

The evaluator model is gemini-3.8-flash, queried with deterministic decoding (temperature 0) under the fixed five-level rubric of Table 4; this model is used exclusively for evaluation and generates none of the content that it scores. The complete evaluation prompt is provided in the Supplementary Material.

Recommendations receiving a relevance score greater than or equal to 4 were regarded as relevant:

$$rel(u, v) = \begin{cases} 1, & Score(u, v) \geq 4 \\ 0, & Score(u, v) < 4 \end{cases} \quad (16)$$

where $Score(u, v)$ refers to either $Score_{LLM}(u, v)$ or $Score_{Human}(u, v)$. Since Level 4 represents "Suitable" and Level 5 represents "Highly suitable", recommendations with an overall relevance score of 4 or higher were treated as relevant for binary evaluation.

Recommendation performance was primarily evaluated at Top-10, because ten recommendations represent a practical recommendation size that users can reasonably inspect. Accordingly, Precision@10, MAP@10, and NDCG@10 were adopted to evaluate recommendation quality and ranking performance. HitRate was additionally reported at Top-3, Top-5, Top-7, and Top-10 to examine whether relevant recommendations appeared early in the ranked list. Mean relevance was also reported as a complementary summary measure based on the 1–5 ordinal relevance ratings. Every reported metric is accompanied by a 95% confidence interval obtained by bootstrap resampling over users with 10,000 replicates and a fixed random seed. Confidence intervals rather than standard deviations are reported because the metrics are bounded and their sampling distributions are not symmetric.

To further examine whether LLM-based evaluation can approximate human judgment, three agreement metrics were employed: Spearman correlation, Mean Absolute Error (MAE), and Quadratic Weighted Kappa (QWK). Spearman correlation measures the consistency of ranking patterns between LLM and human relevance scores. MAE measures the average absolute difference between the two sets of scores. QWK evaluates agreement while accounting for the ordinal nature of the five-level relevance scale. For the QWK analysis, the original integer ratings on the 1–5 scale were used directly for both human and LLM evaluations. Quadratic weighted kappa was then computed on these paired ordinal ratings. The agreement analysis is presented in Section 5.4.

Where a genuine within-unit pairing was available, paired comparisons were analyzed using a paired permutation test with 100,000 sign-flip resamples, a 95% bootstrap confidence interval for the paired difference based on 10,000 replicates, and Holm correction across the four metrics. For automated configuration-level comparisons, the pairing unit was the same synthetic profile evaluated under both configurations. For the human cross-case comparison, the pairing unit was the same participant who evaluated both Case A and Case B. No paired human Core Only versus Core + Adaptive comparison was performed because the human protocol did not include both configurations for the same Case B participants. Cohen's d_z was reported as a standardized effect size; matched-pairs rank-biserial correlations were additionally reported for the automated baseline comparisons. Detailed results are provided in the Supplementary Material.

5.2. Quantitative Recommendation Performance

5.2.1. Performance of Case A: Hobby-to-Volunteer Recommendation

Case A evaluates the performance of the shared core recommendation framework in the hobby-to-volunteer recommendation scenario. Since this scenario does not employ additional domain-aware adaptive signals, it provides a baseline case for examining whether the shared core framework alone can support recommendation across structurally disjoint domains.

The existing-user weights $\alpha = 0.7$, $\beta = 0.2$, and $\gamma = 0.1$ were retained from the previous study and were not re-tuned using the present evaluation profiles. For the cold-start score, five settings of (w_1, w_2) , including the no-transfer setting $(1.0, 0.0)$, were evaluated on the same 50 profiles used for final reporting. Case A selected $(0.9, 0.1)$, whereas Case B selected $(0.6, 0.4)$. Because no held-out set was reserved, these selections and their associated performance estimates are exploratory and may be affected by optimistic selection bias; the complete sensitivity results are therefore reported in Supplementary

Table S5. $K = 15$ was fixed using a similarity-and-coverage diagnostic on the Case B profiles that did not use relevance labels or outcome metrics, but K was not independently validated in Case A.

Case A also serves as the controlled baseline study of this work. To assess the actual contribution of each component, the proposed framework was compared against a panel of baselines under identical experimental conditions: the same candidate pool of 500 target-domain items, the same 50 cold-start profiles, the same Top-10 cut-off, the same five-level relevance rubric, and the same evaluator model. Table 5 reports the comparison with 95% bootstrap confidence intervals.

Several published cross-domain and cold-start approaches require shared or partially aligned users or items, observed target-domain interactions, or bridgeable latent preference spaces. These inputs are unavailable by construction in the structurally disjoint setting considered here. Applying such methods would therefore change the information available to the model rather than provide an input-matched comparison. We consequently focus on baselines that can operate under the same disjoint-domain information constraints, together with a direct LLM ranker as a high-capacity reference comparator.

Table 5. Baseline comparison for Case A under the LLM-based evaluation protocol ($n = 50$ profiles; 95% bootstrap confidence intervals from 10,000 replicates).

Method	Score definition	Precision@10	MAP@10	NDCG@10	HitRate@3
Random	random ordering	0.082 [0.052, 0.116]	0.032 [0.017, 0.050]	0.081 [0.051, 0.115]	0.240 [0.120, 0.360]
Preference only	S_p	0.042 [0.028, 0.056]	0.011 [0.007, 0.016]	0.040 [0.026, 0.055]	0.160 [0.060, 0.260]
Keyword only	S_k	0.088 [0.056, 0.126]	0.039 [0.020, 0.066]	0.091 [0.056, 0.132]	0.240 [0.120, 0.360]
Lexical similarity	TF-IDF cosine	0.258 [0.218, 0.300]	0.132 [0.102, 0.162]	0.264 [0.220, 0.309]	0.680 [0.540, 0.800]
Content hybrid	$0.7S_t + 0.2S_k + 0.1S_p$	0.262 [0.208, 0.320]	0.149 [0.105, 0.197]	0.270 [0.211, 0.330]	0.580 [0.440, 0.720]
Content hybrid with transfer	$0.3S_b + 0.7content$	0.258 [0.208, 0.310]	0.144 [0.107, 0.184]	0.269 [0.213, 0.326]	0.580 [0.440, 0.720]
Semantic similarity only	S_t	0.404 [0.350, 0.458]	0.273 [0.223, 0.325]	0.436 [0.377, 0.494]	0.840 [0.740, 0.940]
Proposed framework	$0.9pct(S_t) + 0.1pct(S_b)$	0.420 [0.380, 0.460]	0.267 [0.228, 0.309]	0.445 [0.401, 0.491]	0.940 [0.860, 1.000]
LLM ranker (reference)	direct selection by an LLM	0.698 [0.636, 0.758]	0.642 [0.570, 0.712]	0.754 [0.696, 0.808]	0.980 [0.940, 1.000]

Table 5 and Figure 4 support three findings. First, the keyword-mapping and preference-compatibility signals, when applied directly to a cold-start user who has no interaction history, perform at or below the random baseline; mixing them into the cold-start score therefore degraded ranking quality rather than improving it, which is why the content hybrid falls well below semantic similarity alone. This is the empirical basis for the revised cold-start scoring function described in Section 3.3. Second, the revised framework significantly outperforms the earlier content-hybrid formulation with transfer (paired difference +0.162 in Precision@10, 95% CI [0.114, 0.214], Holm-adjusted $p < 0.001$, Cohen's $d_z = 0.90$). Third, the revised framework is not statistically distinguishable from the semantic-similarity-only baseline in this case study (+0.016 in Precision@10, Holm-adjusted $p = 1.00$); we return to the interpretation of this result in Section 5.2.5.

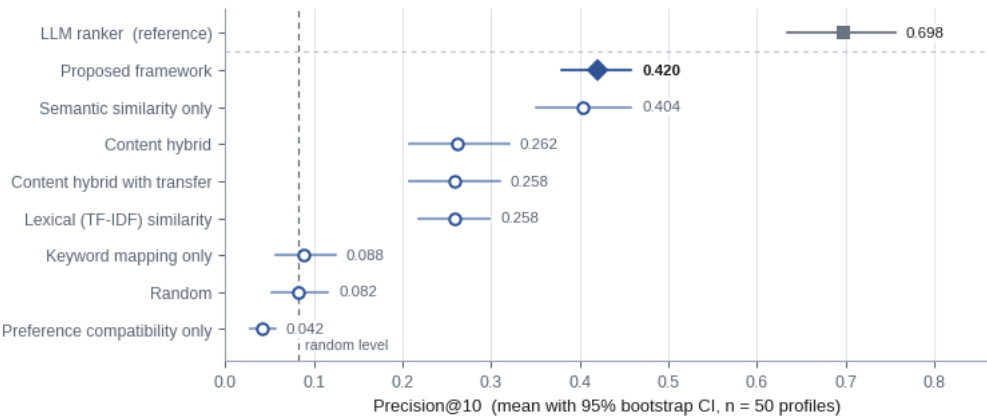


Figure 4. Baseline comparison for Case A. Mean Precision@10 with 95% bootstrap confidence intervals for the nine compared methods, ordered by performance. The language model ranker is shown separately as a high-capacity reference comparator rather than as a directly scalable scoring-based baseline.

Table 6. Case A results under the automated and the human protocol.

Metric	Automated protocol (proposed framework)	Human protocol (content hybrid with transfer)
Precision@10	0.420 [0.380, 0.460]	0.640 [0.557, 0.717]
MAP@10	0.267 [0.228, 0.309]	0.723 [0.636, 0.805]
NDCG@10	0.445 [0.401, 0.491]	0.832 [0.765, 0.892]
HitRate@3	0.940 [0.860, 1.000]	0.867 [0.733, 0.967]
Mean relevance	3.21 [3.13, 3.30]	3.77 [3.57, 3.97]

In Table 6 the automated protocol is applied to the proposed framework on $n = 50$ synthetic profiles and the human protocol to the content-based configuration listed in Table 5 as Content hybrid with transfer on $n = 30$ participants, each with 95% bootstrap confidence intervals. The two columns differ in the profile set and in the recommendation configuration, so they are not a comparison of the same system; they appear together because both were scored under the identical relevance question and five-point scale of Table 4.

Under the automated protocol, the framework attained $\text{Precision@10} = 0.420$ and $\text{HitRate@3} = 0.940$; under the human protocol, $\text{Precision@10} = 0.640$. Because the protocols used different profile sets, the numerical difference should not be interpreted as an evaluator effect. Both protocols nevertheless yielded high HitRate@3 , indicating that nearly every evaluated profile received at least one relevant item near the top of the list.

The remaining gap to a perfect Precision@10 reflects the inherent difficulty of this domain pair. Volunteer preferences must be inferred indirectly from hobby-related interests, and hobbies do not always correspond to specific volunteer activities; hobbies with clear social or activity-based implications are easier to connect with volunteer opportunities than hobbies whose semantic relationship to volunteering is weak. The baseline comparison in Table 5 makes this concrete: even a language model that inspects the entire candidate set and selects by reasoning reaches only 0.698 on the same profiles and the same candidate pool, which suggests that a substantial part of the residual error is a property of the task rather than of the scoring function.

5.2.2. Performance of Case B: Consumption-to-Travel Recommendation

Case B evaluates the overall performance of the proposed framework in the consumption-to-travel recommendation scenario. Unlike Case A, this scenario activates the domain-

aware adaptive layer, so that lifestyle, destination-attribute and budget signals contribute to the score for those users who supplied the corresponding preference. Table 7 reports Case B performance for the 50 synthetic profiles and the 30 human participants described in Section 5.1.

Table 7. Case B results under the automated and the human protocol.

Metric	Automated protocol (proposed framework)	Human protocol (content hybrid with transfer)
Precision@10	0.630 [0.556, 0.700]	0.653 [0.597, 0.710]
MAP@10	0.570 [0.493, 0.646]	0.724 [0.661, 0.786]
NDCG@10	0.694 [0.628, 0.757]	0.839 [0.793, 0.885]
HitRate@3	0.980 [0.940, 1.000]	0.967 [0.900, 1.000]
Mean relevance	3.69 [3.54, 3.83]	3.85 [3.70, 4.02]

Table 7 follows the same convention as Table 6: the automated protocol is applied to the proposed framework on $n = 50$ synthetic profiles and the human protocol to the content-based configuration on $n = 30$ participants, each with 95% bootstrap confidence intervals.

As shown in Table 7, the proposed framework achieved strong recommendation performance under both evaluation settings. Precision@10 reached 0.630 under the automated protocol and 0.653 under the human protocol, indicating that 63.0% and 65.3% of the Top-10 recommendations, respectively, were classified as relevant. MAP@10 was 0.570 and 0.724 and NDCG@10 was 0.694 and 0.839 under the automated and human protocols. These values indicate that relevant destinations were not only frequently included in the Top-10 lists but were also generally positioned near the top of the rankings. HitRate@3 was high under both protocols, 0.980 and 0.967 respectively, so almost every evaluated user received at least one relevant destination within the first three recommendations. Because the same 30 participants evaluated one Top-10 recommendation list in each case study, Case A and Case B were additionally compared at the participant level. The Case B minus Case A differences were +0.013 for Precision@10, +0.001 for MAP@10, +0.008 for NDCG@10, and +0.100 for HitRate@3. None of these differences was statistically significant after Holm correction (all adjusted $p = 1.00$). The corresponding confidence intervals and effect sizes are reported in Supplementary Table S11a. This analysis is a cross-case comparison and should not be interpreted as an estimate of the adaptive-layer effect because the two cases differ in both domain and model configuration.

5.2.3. Contribution of the Domain-Aware Adaptive Layer

All configuration-level analyses in this subsection were conducted under the automated evaluation protocol using the same 50 synthetic Case B profiles. The human-evaluation protocol did not include a second Case B recommendation list generated with the adaptive layer disabled and therefore does not support a paired human Core Only versus Core + Adaptive comparison.

Because the revised adaptive layer activates each domain-specific signal only for users who supply the corresponding preference, the 50 evaluation profiles of Case B cover all eight combinations of the three signal-activation indicators, with all eight activation patterns represented. This allows the contribution of the layer to be examined at the level of the individual signals rather than only as a whole, which is what the aggregate comparison reported in the submitted version could not do.

Table 8 groups the profiles by the number of adaptive signals that are active. Precision@10 rises from 0.300 with no signal to 0.669 with one and 0.792 with two, and does

not increase further with all three (0.783); the two- and three-signal groups are not distinguishable at this sample size. Aggregating over the groups in which each individual signal is active provides a descriptive indication of where the observed differences are concentrated.

Table 8. Case B performance grouped by the number of active adaptive signals.

Active signals	Patterns	Profiles	Precision@10	MAP@10	NDCG@10	Control (LLM ranker)	Gap
0 (core only)	CORE	13	0.300 [0.231, 0.369]	0.232 [0.159, 0.306]	0.397 [0.308, 0.482]	0.569 [0.469, 0.669]	−0.269
1	L 5, C 4, B 4	13	0.669 [0.531, 0.800]	0.610 [0.467, 0.752]	0.730 [0.622, 0.836]	0.808 [0.700, 0.908]	−0.138
2	LC 4, LB 4, CB4	12	0.792 [0.708, 0.875]	0.743 [0.635, 0.847]	0.841 [0.771, 0.907]	0.867 [0.767, 0.950]	−0.075
3	LCB 12	12	0.783 [0.725, 0.842]	0.721 [0.646, 0.795]	0.829 [0.780, 0.878]	0.883 [0.808, 0.950]	−0.100

All values in Table 8 are means over the profiles in each group, with 95% bootstrap confidence intervals from 10,000 replicates. The control column reports the language model ranker evaluated on the same profiles, and the gap column is the paired difference between the proposed framework and that control.

L denotes lifestyle-trait matching, C destination-attribute matching and B budget compatibility; the Patterns column gives the number of profiles carrying each pattern. All group means are computed over users rather than over pooled recommendation instances, so that the profile-weighted average of the four groups reproduces the overall values of Tables 7 and 9 exactly (0.630 for the proposed framework and 0.778 for the control).

Because these groups are defined by which preferences a user chose to supply rather than by randomized assignment, the gradient could in principle reflect the fact that users who express more preferences have more specific profiles and are therefore easier to serve. We tested this directly using the language model ranker as a control: that baseline receives the same profile fields as free text but contains no adaptive scoring layer, so any gradient it shows reflects profile specificity rather than the module under examination. The control confirms part of this concern and we report it: the baseline without an adaptive layer also improves across the groups, from 0.569 to 0.883 in Precision@10.

The decisive quantity is therefore not the gradient itself but the difference between the two curves. The gap between the proposed framework and the control narrows as adaptive signals become available, from −0.269 with no signal, to −0.138 with one, to −0.075 with two, before widening slightly to −0.100 with all three. Approximately 0.17 to 0.19 of Precision@10 is thus recovered relative to the difficulty baseline, which we interpret as the contribution of the explicit adaptive layer. We state plainly that this is the strongest evidence obtainable without a randomized design, that the group sizes are small, and that a controlled component-level ablation evaluating the same users with each signal switched on and off remains the appropriate confirmatory design.

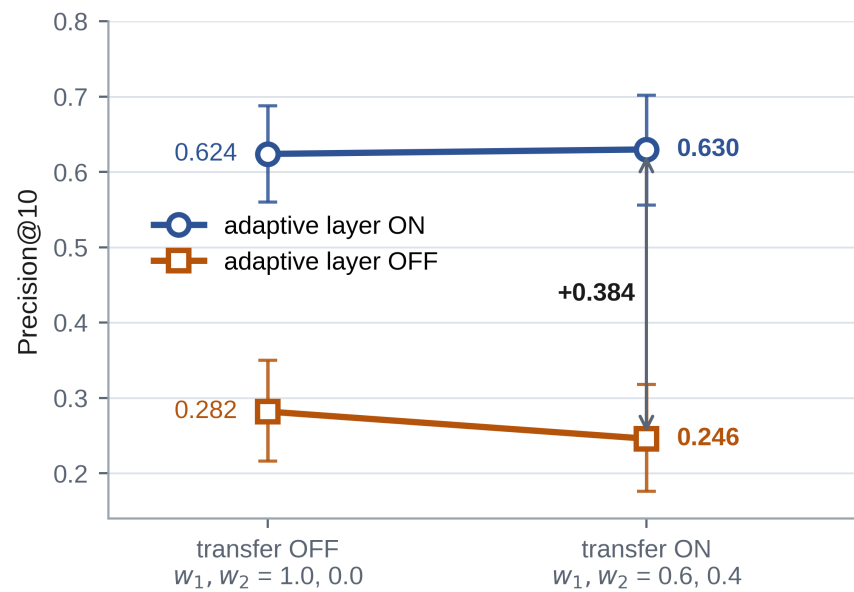


Figure 5. Two-by-two factorial analysis of the Case B framework. The horizontal axis contrasts the two settings of the cold-start weights, (w_1, w_2) = (1.0, 0.0) with the transfer term switched off and (0.6, 0.4) with it active; the two lines contrast the adaptive layer switched on and off. Points are mean Precision@10 over the 50 evaluation profiles and error bars are 95% bootstrap confidence intervals. The vertical arrow marks the contribution of the adaptive layer at the selected weight setting. The near-horizontal slope of both lines shows that the transfer term does not change the ranking appreciably, while the vertical separation between them shows the effect of the adaptive layer.

The corresponding activation-pattern analysis and control curve are reported in Supplementary Figure S6. The configuration-level contrast shown in Figure 5 was conducted under the automated protocol by evaluating the same 50 synthetic Case B profiles with the adaptive layer enabled and disabled while holding the core-score formulation fixed. This within-profile contrast was analyzed using the paired procedure described in Section 5.1. By contrast, the activation-pattern summaries in Table 8 compare different groups of profiles and are therefore interpreted as descriptive groupwise results rather than as paired comparisons. No corresponding within-participant configuration-level comparison was available under the human protocol.

Taken together, these analyses support the conclusion that the domain-aware adaptive layer contributes beyond the shared core layer, while making explicit that part of the raw group difference is attributable to profile specificity rather than to the layer itself.

5.2.4. Comparison with a Language Model Used Directly as a Ranker

To provide a demanding external comparator rather than only internal ablation, we implemented a baseline in which a language model receives the raw source-domain profile and the raw target-domain candidate set and selects the Top-10 directly. It is given no scoring signals, no similarity computation and no instruction that is unavailable to the other baselines; candidates carry opaque identifiers, their order is shuffled once with a fixed seed, and exactly one call is issued per user with no ensembling or repeated querying. Table 9 reports the comparison in both case studies.

Table 9. Proposed framework compared with a language model used directly as a ranker.

Case study	Metric	Proposed LLM ranker	Paired difference	Holm-adjusted p	Cohen's dz
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Case A	Precision@10	0.420	0.698	-0.278	< 0.001	-1.06
Case A	NDCG@10	0.445	0.754	-0.309	< 0.001	-1.15
Case A	HitRate@3	0.940	0.980	-0.040	0.633 (n.s.)	-0.14
Case B	Precision@10	0.630	0.778	-0.148	< 0.001	-0.69
Case B	NDCG@10	0.694	0.863	-0.170	< 0.001	-0.73
Case B	HitRate@3	0.980	0.980	0.000	1.000 (n.s.)	0.00

Table 9 reports the LLM-based evaluation protocol with $n = 50$ profiles per case study, analyzed with a paired permutation test and Holm correction across the four metrics. MAP@10 is omitted because it follows the same pattern as NDCG@10.

The LLM ranker achieved higher Precision@10 and NDCG@10 in both cases, whereas HitRate@3 did not differ significantly. It should be treated as a high-capacity reference comparator: it processes the entire candidate set for each user, does not scale like item-wise scoring, and neither performs similar-user transfer nor yields decomposable component scores.

5.2.5. Comparative Discussion

The results of Cases A and B demonstrate that the proposed framework can support recommendation across structurally disjoint domains. However, recommendation performance differed according to the semantic relationship between the source and target domains and the amount of domain-specific information available for recommendation.

Case A represented a relatively challenging setting because hobby preferences and volunteer activities are connected indirectly. Although the shared core framework consistently retrieved at least one relevant item within the Top-10 list, the moderate Precision@10 values indicate that some recommendations were only partially aligned with users' interests. This result reflects the difficulty of transferring preferences between domains with substantially different purposes and behavioral contexts.

In contrast, Case B achieved higher overall ranking performance. Consumption behavior provides broader information about lifestyle orientation and purchasing tendencies, while the travel dataset contains detailed destination attributes. These characteristics enable more explicit matching between user-related signals and target-domain items.

The two-by-two factorial analysis in Case B separates the two components of the revised framework. Crossing the transfer term with the adaptive layer gives four conditions, shown in Figure 5. The domain-aware adaptive layer accounts for a large and consistent improvement: averaged over the two transfer settings it raises Precision@10 by 0.363 (paired permutation test over the 50 profiles, $p < 0.001$, Cohen's $d_z = 1.04$), and the corresponding simple effect at the selected weight setting is +0.384 (Holm-adjusted $p < 0.001$, $d_z = 1.01$). The behaviour-transfer term does not produce a statistically distinguishable change in the ranking: comparing $(w_1, w_2) = (0.6, 0.4)$ with $(1.0, 0.0)$ under the adaptive layer gives +0.006 Precision@10 (Holm-adjusted $p = 1.00$, uncorrected $p = 0.83$, $d_z = 0.05$), and the corresponding comparison in Case A is likewise not significant (+0.016, Holm-adjusted $p = 1.00$). The interaction between the two factors is +0.042 and does not reach the significance level ($p = 0.098$).

The factorial results clarify the empirical role of the transferred signal. The transfer term introduces information from the existing-user population into the cold-start ranking; however, it did not produce a statistically detectable incremental ranking improvement in either evaluated domain pair. One possible explanation is that neighbor retrieval and direct user-item matching use the same sentence-embedding space, making the transferred ordering similar to a smoothed version of the direct semantic ordering. In addition, averaging over $K = 15$ neighbors may attenuate between-user variation. We therefore retain behavior transfer as a structural mechanism for incorporating existing-user information, but not as an empirically demonstrated source of ranking improvement in the

present experiments. Evaluating whether transfer becomes beneficial in domain pairs with stronger behavioral correspondence remains an important direction for future work.

5.3. Qualitative Recommendation Example

5.3.1. Case A: Hobby-to-Volunteer Examples

To examine the behavior of the shared core framework qualitatively, two of the fifty synthetic cold-start personas were analyzed in detail. Both examples were obtained under the final Case A configuration reported in Section 5.2 and correspond to the Top-1 recommendation. The personas were generated and the explanations produced in Korean, and both are presented here in translation. Table 10 presents the first example, based on a craft- and baking-oriented profile.

Table 10. Case A qualitative example 1: a craft- and baking-oriented cold-start user.

Example 1 dexterity-oriented everyday carer
user profile:
On weekends I like to turn on the oven and bake cookies or muffins to share with my family and close friends. Recently I have taken up knitting small items, which I am teaching myself from videos. Sometimes I sit alone in a neighborhood café and spend a quiet hour organizing my recipe notes. I find it burdensome to move around vigorously in a large group, or to join improvised outdoor group games.
The Top-1 recommended volunteer activities are presented below:
© [Hagye Day Activity Centre for the Disabled] Volunteers wanted (assistance with the baking program)
LLM-generated Top-1 Explanation
1. Since the user enjoys baking cookies and muffins and sharing them with family and friends, assisting a program run in a bakery room lets that familiar baking experience continue as a meaningful in-person activity
2. Because the user prefers to concentrate quietly indoors on knitting or on organizing recipes, participating on the fixed biweekly Thursday schedule at the bakery room of Hagye Community Welfare Centre is likely to suit them and to keep the demands low.
3. In helping adults with severe brain-lesion disabilities through the baking program, the user can find a reward beyond the pleasure of making something — being close at hand while someone else takes part and completes their own work.

This example illustrates two aspects of the revised formulation. First, only the semantic term and the transferred behavioral term are active in the score, since the keyword-mapping and preference-compatibility terms are not computed for cold-start users, in accordance with Equation (9). Second, the recommended item is not classified as a baking activity in the volunteer platform taxonomy; its declared field is educational support within facility volunteering. The correspondence between the user profile and the item is therefore established through the semantic representation of the item description and through the behavior of similar existing users, rather than through any categorical relation

between the two domains. Such an item would not be retrieved by a keyword-based mapping.

Table 11. Case A qualitative example 2: an outdoor- and sports-oriented cold-start user.

Example 2 outdoor-activity enthusiast
user profile:
Every weekend I head out to a nearby mountain or riverside for hiking or cycling, which also keeps me fit. I recently joined a running crew and discovered how enjoyable it is to run with other people; I feel energized by moving my body outdoors. Sorting out camping gear and planning trips is part of the hobby too. I tire quickly of sitting indoors for long stretches on detailed work, or of static lecture-format activities.
The Top-1 recommended volunteer activities are presented below:
© [Jungwon Senior Welfare Centre] 2025 Senior Citizen Support Team joint external activity

LLM-generated Top-1 Explanation

1. Since the user enjoys moving outdoors with other people — weekend hiking and cycling, and the running crew — accompanying an outdoor excursion to Yeosu and helping it run is likely to feel far more natural and energising than a static indoor placement.
2. The role involves helping older participants take part in an outing comfortably and enjoyably in person, which matches the user's wish to contribute directly to the local community and to interact with others.
3. It is a single-day, non-recurring activity, so it fits the kind of schedule the user already keeps for running and camping, and it offers the experience of accompanying older adults on an outing and supporting their leisure time while meeting new people.

As shown in Tables 10 and 11, the shared core framework distinguishes cold-start users according to the semantics of their stated interests. A solitary indoor craft profile is matched to a bakery-room placement, whereas an outdoor group-sports profile is matched to a single-day excursion, and the two recommendation sets do not overlap, although both users were processed by an identical pipeline under identical settings. The generated explanations also refer to different elements of each profile, including the activities that each user stated a preference to avoid.

The two examples further indicate a limitation that is not apparent from the aggregate metrics. Neither recommendation falls within the district stated as preferred by the user, and among the lower-ranked items for both users the evaluator assigned reduced relevance to placements that were semantically appropriate but geographically difficult to attend. Region is represented only through the item text and is not applied as a constraint, so semantically relevant but practically unreachable items are not excluded. The introduction of region as an explicit constraint, or as an additional gated signal in the sense of Section 3.4, constitutes a direct extension of the present framework and was not evaluated in this study.

5.3.2. Case B: Consumption-to-Travel Examples

The Case B examples are presented as a contrasting pair rather than as two instances of the same condition, since the property under examination is the contribution of the adaptive layer. Under the indicator-gated score of Equation (11), an adaptive signal enters the final score only when that signal can be induced from the consumption record of the user concerned, so the set of active signals $A(x)$ varies across users. Table 12 therefore presents a user for whom no adaptive signal is available, so that the recommendation is produced by the shared core alone, and Table 13 presents a user for whom all three signals are available. Both examples were obtained under the final Case B configuration reported in Section 5.2, and both explanations were generated in Korean and are presented in translation.

Table 12. Case B qualitative example 1: a consumer profile with no active adaptive signal.

Example1. home-office equipment buyer
User Profile:
This shopper has built up a small but purposeful home-office setup, favoring gear that supports long hours at a desk. Recent purchases include a PC headset for daily video calls, an HDMI cable to drive an external monitor in a dual-screen workflow, and a compact USB hub to manage peripherals without cluttering the desk. Product research focuses on sound clarity, latency and build durability, with a consistent preference for devices that work reliably out of the box and for well-reviewed mid-to-premium products over the cheapest option.
The Top-1 recommended travel destinations are presented below:
● Helsinki, Finland
Adaptive Interpretation
● Lifestyle Trait ($S_l(n, v)$): not induced — signal inactive
● City Attributes ($S_c(n, v)$): not specified — signal inactive
● Budget Preference ($S_b(n, v)$): not specified — signal inactive
LLM-generated Top-1 Explanation
1. Given how carefully the user weighs performance, durability and a clean, uncluttered experience, the orderly atmosphere of Helsinki and its pronounced modern design are a good fit
2. Just as the user manages cables and devices efficiently at the desk, a preference for convenience and finish over complexity sits naturally with Helsinki's spare contemporary sensibility.
3. Because the user values a comfortable environment after long working hours, Helsinki — with its clean air and welcoming cafés — offers the chance to rest quietly while taking in a design-led cityscape.

Table 12 corresponds to the condition in which the adaptive layer is inactive, and the user shown is the same user illustrated in Figure 3. The consumption record supported no lifestyle keyword, no destination attribute and no budget level, so $A(x)$ is empty and

the final score reduces to the normalized core score. The recommendation is therefore determined by the semantic similarity between the consumption profile and the destination descriptions together with the aggregated travel behavior of similar existing users. The generated explanation is formulated in terms of atmosphere and design rather than in terms of a stated travel preference, and introduces no attribute that is absent from the profile.

Table 13. Case B qualitative example 2: a consumer profile with all three adaptive signals active.

Example2. audio-visual and photography equipment buyer
User Profile: This shopper has been steadily upgrading a home entertainment and workspace setup, favoring gear that delivers crisp visuals and rich sound. Recent purchases include high-speed HDMI cables, a compact soundbar to replace the television's built-in audio, and a sturdy tripod for recording product reviews and tutorials at the desk. The buyer reads reviews closely, comparing bandwidth specifications and shielding quality before committing, and gravitates toward recognized brands known for reliable connectors and long-term durability.
The Top-1 recommended travel destinations are presented below: ⊗ New York City, United States
Adaptive Interpretation ⊗ Lifestyle Trait ($S_l(n, v)$): urban experiences, sensory ambiance, vibrant atmosphere, eclectic blend ⊗ City Attributes ($S_c(n, v)$): North America; weekend or short trip; mild climate ⊗ Budget Preference ($S_b(n, v)$): Premium
LLM-generated Top-1 Explanation 1. Because the user cares about sharp images, rich sound and a reliably assembled setup, the intense audiovisual character of New York's high-rise skyline and busy streets is a particularly close match. 2. Given that the user compares reviews closely and prefers well-resolved choices, the layered appeal of New York — where many cultures and identities meet in one place, so experiences can be chosen in fine detail — is well suited to them. 3. As a premium weekend or short trip in North America, it is a good setting for exploring an energetic city center in mild weather and concentrating on its sensory streetscapes and lively atmosphere.

A comparison of Tables 12 and 13 shows the effect of the indicator gating. The two users have a broadly similar consumer-electronics purchase history, and in both cases the shared core retrieves a plausible destination without the adaptive layer. For the second user, however, lifestyle, destination-attribute and budget evidence could be induced from the consumption record, so the corresponding gated terms enter the score and reorder the candidate list. A corresponding difference appears in the generated explanations: the explanation in Table 13 refers to the region, trip duration, climate and budget level derived from the user profile, whereas the explanation in Table 12 has no such evidence available

and does not refer to these attributes. The adaptive layer therefore does not replace the shared core, but refines its output for those users whose source-domain records provide sufficient evidence, and leaves it unchanged otherwise.

Since the number of gated terms depends on $A(x)$, the final score is a within-user ranking quantity whose scale varies with the number of active signals. It is used to order candidate items for a given user and is not intended for comparison between users with different signal patterns. The quantitative counterpart to these examples, including the separate contributions of the adaptive layer and of the transfer term, is reported in Section 5.2.

5.4. Agreement between LLM-Based and Human Evaluation

In addition to recommendation performance, the agreement between LLM-based and human evaluation was examined to assess whether automated relevance judgments were consistent with human assessments. The analysis was based on the paired recommendation instances described in Section 5.1, using Spearman correlation, Mean Absolute Error (MAE), and Quadratic Weighted Kappa (QWK). Table 14 summarizes the results for Cases A and B.

The Spearman correlation was 0.3632 in Case A and 0.4271 in Case B, indicating positive but weak-to-moderate associations between LLM-based and human relevance judgments. The corresponding MAE values were 0.9446 and 0.7771, respectively, indicating somewhat closer agreement between the two evaluators in Case B.

The QWK values were 0.2167 for Case A and 0.3265 for Case B. According to the commonly used interpretation proposed by Landis and Koch [56], both values fall within the fair-agreement range, although the Case A value is close to its lower boundary. Thus, the ordinal agreement between LLM-based and human evaluation was limited but consistently positive across both scenarios.

Table 14. Agreement between LLM-based and human evaluation for Case A and Case B.

Metric	Case A	Case B
Spearman Correlation	0.3632	0.4271
MAE	0.9446	0.7771
Quadratic Weighted Kappa	0.2167	0.3265

Overall, the results suggest that LLM-based evaluation captures some aspects of human relevance judgment but does not closely reproduce human assessments. The somewhat higher correlation, lower MAE, and higher QWK observed in Case B suggest greater agreement in the consumption-to-travel scenario, where consumption-derived preferences and structured destination attributes may provide clearer cross-domain correspondence. Therefore, LLM-based evaluation is better interpreted as a scalable complementary assessment method rather than a substitute for human evaluation.

Because this study relies on automated relevance assessment for the large-scale comparisons, we state explicitly the risks this carries and the controls applied to each. Self-preference, the tendency of an evaluator to favour output resembling its own generations, is addressed by a three-way model separation in which profiles, explanations and evaluations are produced by three different models, so that no model evaluates its own output and the evaluator is used for no other purpose. Position bias is addressed for the language model ranker by shuffling the candidate order once with a fixed seed; for the human evaluators the order was not randomized, which we state as a limitation. Verbosity bias is limited by constraining explanations to exactly three sentences with a fixed template. An

exploratory Case A control analysis did not detect a statistically significant difference between automated relevance scores obtained with and without the generated explanation. However, this result does not establish equivalence and does not address the possible influence of explanations on human judgements or in Case B. Non-determinism is limited by deterministic decoding with a fixed rubric and a fixed seed, although we do not claim that inference is formally deterministic. Divergence from human judgment is reported quantitatively using three complementary agreement metrics rather than a single correlation.

On this basis we regard automated relevance evaluation as defensible in this setting under four conditions: the evaluator is a different model from the models used for profile and explanation generation; the rubric is fixed, explicit and identical to the one given to human evaluators; agreement with human judgment is measured and reported rather than assumed; and the automated scores are used for relative comparison between methods under identical conditions rather than as absolute estimates of quality. Where these conditions do not hold we would not regard automated evaluation as substituting for human assessment, and we do not claim that it does here.

5.5. Exploratory Control Analysis of Explanation-Associated Effects

To examine whether the generated explanation influenced automated relevance scoring, we conducted an exploratory paired control analysis in Case A. The same recommendation instances from the 50 evaluation profiles were assessed twice under otherwise identical settings: once with the generated explanation included and once with the explanation omitted. The per-profile mean relevance score across the Top-10 recommendations was used as the paired unit of analysis.

A two-sided Wilcoxon signed-rank test detected no statistically significant difference between the explanation-present and explanation-absent conditions ($n = 50$ profiles, $W = 447$, $p = 0.955$). Thus, no explanation-associated difference was statistically detected in the automated Case A evaluation.

This result should be interpreted within the scope of the analysis. The absence of a statistically significant difference does not establish statistical equivalence or demonstrate that explanations cannot influence relevance judgements. The analysis was exploratory and was limited to the automated evaluator in Case A. No corresponding explanation-absent condition was implemented for the human evaluation or for Case B. The possible influence of explanations on human judgements therefore remains unresolved.

6. Conclusion and Future Work

This study proposed an adaptive cross-domain recommendation framework for heterogeneous environments in which source and target domains are structurally disjoint. Unlike conventional cross-domain recommendation methods that depend on shared users, shared items, aligned attributes, or comparable interaction structures, the proposed framework transfers recommendation knowledge through textual semantic representation, hybrid scoring, similar-user retrieval, and behavior transfer.

The framework consists of two complementary components: a shared core recommendation layer and a domain-aware adaptive module. The shared core layer provides a reusable recommendation mechanism based on semantic similarity, keyword-based mapping, preference compatibility, and similar-user knowledge transfer. The adaptive module extends this common architecture by incorporating additional signals that reflect the characteristics of a particular recommendation scenario.

The proposed framework was evaluated in two heterogeneous cross-domain scenarios. In Case A, hobby-domain user information was used to recommend volunteer activi-

ties. The results showed that the shared core framework could generate relevant recommendations even without shared users, shared items, or domain-specific adaptive signals. Although Precision@10 and MAP@10 were moderate, the high NDCG@10 and HitRate values indicated that relevant volunteer activities were generally positioned near the top of the recommendation lists. The revised experiments add a controlled baseline comparison in this case study. The proposed framework significantly outperformed the two hybrid formulations that directly included keyword and preference signals but was not statistically distinguishable from semantic similarity alone. The Case B factorial analysis likewise found no statistically detectable incremental ranking improvement from behavior transfer, whereas the domain-aware adaptive layer produced a large improvement. Accordingly, behavior transfer should be interpreted as a mechanism for incorporating information from the existing-user population rather than as a demonstrated source of performance improvement in the two evaluated domain pairs.

In Case B, consumption behavior was used to recommend travel destinations. The framework with the adaptive layer active achieved strong performance under both the automated and the human evaluation protocol, with Precision@10 of 0.630 and 0.653 respectively and HitRate@3 above 0.96 in both cases.

The contribution of the adaptive layer was examined at the level of the individual signals rather than as a whole. Because each domain-specific signal is activated only for users who supply the corresponding preference, the evaluation profiles form a factorial design over the three signals, and performance increases with the number of active signals before saturating. Using a baseline without an adaptive layer as a control, roughly 0.17 to 0.19 of Precision@10 is attributable to the layer itself once differences in profile specificity are accounted for, with lifestyle-trait and destination-attribute matching contributing more than budget compatibility.

A language model used directly as a ranker outperforms the proposed framework on precision and ranking quality in both case studies. We report this without qualification and position the two as occupying different operating regimes: the ranker inspects the entire candidate set within a single context window and is therefore better read as a reference upper bound, the two are statistically indistinguishable on early-rank quality, and the ranker neither scales to large candidate sets nor implements the transfer mechanism studied here.

Taken together, the experimental results support the main design principle of the proposed framework. The shared core layer provides a reusable recommendation mechanism across structurally disjoint domains, while the domain-aware adaptive module further improves recommendation relevance and ranking quality when meaningful domain-specific information is available. The framework therefore balances architectural generality with scenario-specific adaptability.

This study also examined the supporting role of LLMs in recommendation explanation and evaluation. The LLM-based explanation module generated natural-language rationales grounded in user profiles, target-domain item information, and recommendation evidence. In addition, LLM-based evaluation was used as a scalable relevance-assessment method in settings where explicit cross-domain ground-truth interactions were unavailable. The agreement analysis showed positive correlations between LLM-based and human evaluation, with limited-to-fair ordinal agreement. These findings suggest that LLM-based evaluation can serve as a useful complementary assessment method, but should not be regarded as a replacement for human judgment.

Several limitations remain. First, the framework was evaluated using only two cross-domain scenarios, and its generalizability should be validated across a broader range of source-target domain pairs; this is particularly important because in neither of the two pairs examined did the transferred signal produce a measurable change in the ranking.

Second, the two weights of the cold-start score were selected on the same profiles used for the reported results, without a held-out split; no post-hoc selection or exclusion of evaluation data was performed at any stage, the same fifty profiles were used for every weight setting, and all five settings are reported. The factorial analysis limits the consequence of this choice, since the settings are not statistically distinguishable after correction. Third, the analysis of the adaptive signals is based on groups defined by which preferences a user chose to supply rather than by randomized assignment, and the group sizes are small, so a controlled component-level ablation remains necessary for a confirmatory ranking of the three signals. Fourth, the human evaluation involved 30 participants, and items were presented in descending order of the model score rather than in randomized order, which may induce a position effect. Fifth, the human protocol did not include paired Core Only and Core + Adaptive recommendation lists for the same Case B participants. Consequently, the configuration-level contribution of the adaptive layer was evaluated only under the automated protocol. A randomized or counterbalanced human ablation in which the same participants evaluate both configurations remains necessary. Finally, the reliability of LLM-based evaluation requires further investigation using larger and more diverse human evaluation samples.

Future work will focus on five concrete directions, each tied to a component that remains manual or under-determined in the present study. First, automatic rule generation will be promoted from an auxiliary step to a validated module by removing the remaining author-review stage, validating generated rules automatically against held-out data, and quantifying how sensitive recommendation quality is to rule quality. Second, the grid search over the two core weights will be replaced by learned weighting as a function of measurable properties of the domain pair, such as the distributional distance between the source and target embeddings, so that a new domain pair receives an inferred rather than a tuned weight. Third, the binary activation of the adaptive signals will be generalized to a continuous per-user weighting learned from interaction feedback, which would subsume both the current indicator scheme and the fixed-weight scheme it replaced and would allow the sub-component contributions to be estimated within a single model. Fourth, a held-out evaluation protocol, or nested cross-validation over user profiles, will be established as a design requirement for subsequent work in this setting. Fifth, we will conduct a larger-scale human evaluation incorporating a randomized or counterbalanced explanation-present versus explanation-absent condition, using identical recommendation instances and paired judgements to distinguish item relevance from the influence of explanation quality.

Supplementary Materials: The following supporting information is available online: Supplementary Methods S1–S6; Tables S1–S11 and S11a; Figures S1–S6. The Supplementary Material reports the generated keyword-mapping rules, neighbor-size diagnostic, complete weight-sensitivity results, LLM configurations and prompts, and paired statistical analyses with standardized effect sizes.

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